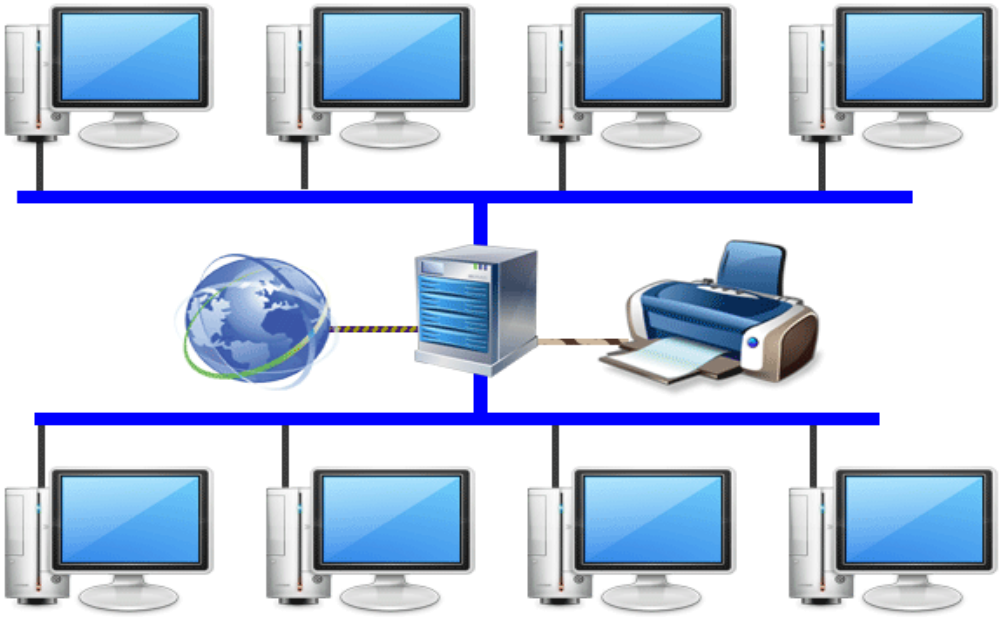
Introduction to Computer Network

- A system of interconnected computers and computerized peripherals such as printers is called computer network
- Two computers are said to be interconnected if they are able to exchange information



Introduction to Computer Network

- Large number of separate but interconnected computers
- A node can be a computer, telephone, or any other device capable of sending and receiving messages.
- The communication medium is the physical path by which message travels from sender to receiver. Example media include fiber-optic cable, copper wire, or air carrying radio waves.
- The purpose of a network is to share resources
- A resource may be
  - A file
  - A folder
  - A printer
  - A disk drive
  - Or just about anything else that exists on a computer.

Goals of Computer Networks

- "Resource sharing" - to make all programs, data and equipment available to anyone on the network without the regard to the physical location of the resource and the user.
- To provide high “reliability” by having alternative sources of supply. For example, all files could be replicated on more than one, so if one of them is unavailable, the other copies could be available.
- Cost saving
- Increase the systems performance
- Powerful communication medium

Advantages / costs of Networking

- Connectivity and Communication, Data Sharing, Hardware Sharing Internet Access, Internet Access Sharing, Data Security and Management, Performance Enhancement and Balancing, Entertainment
- Disadvantages (Costs) of Networking
  - Network Hardware, Software and Setup Costs
  - Hardware and Software Management and Administration Costs
  - Undesirable Sharing
  - Illegal or Undesirable Behavior
  - Data Security Concerns

Uses of Computer Networks

Why people are interested in computer networks and what they can be used for?

- Business Applications
- Home applications
- Mobile user

- Social Issues

Uses of Computer Networks

Business applications

- Information sharing in company
- The computers and employees may be scattered over dozens of offices and plants in many countries
- Group of workers – sharing a common printer (high volume networked printer)
- Company information database in powerful central computer (server) accessed by employees simpler machines (clients) on their desks
- Another goal apart from information sharing on computers is Powerful Communication medium for people
- email (Electronic mail), IP telephony or Voice over IP (VoIP), Video conferencing, Desktop sharing

Uses of Computer Networks

Home applications

1. Interactions between a person and a remote database full of information
   - Client-Server model
   - Peer-to-Peer communication
2. Person-to-person communication
3. Electronic commerce
4. Entertainment
5. Ubiquitous computing

Uses of Computer Networks
Mobile Users

- Mobile computers, such as laptop and handheld computers are being used more than desktop computers as they can be carried and used wherever we go
- Mobile - Wireless **hotspots –** net access by laptops instead plugged wire connection
- Wireless networks important for taxis, trucks, delivery vehicles, repairing persons to keep in touch with home base
- Its own wireless network for military is important
- Smart phones
- GPS equipped mobile phones
- M-commerce (Mobile commerce) – payment authorization through verification code via mobile phone

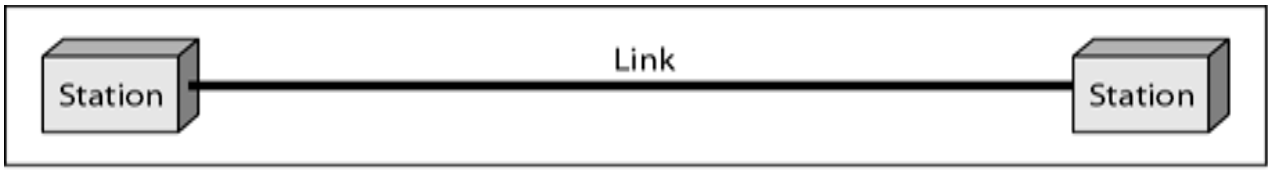
Uses of Computer Networks
Social Issues

- Computer networks allow ordinary citizens to distribute and view content in ways that were not previously possible
- Social networks, message boards, content sharing sites, and a host of other applications allow people to share their views with like-minded individuals.
- As long as the subjects are restricted to technical topics or hobbies, not too many problems will arise.
- The trouble comes with topics that people actually care about, like politics, religion, or sex.
- Views that are publicly posted may be deeply offensive to some people.
- Controversial contents ?? Pirated music and movies?
- Different countries different and conflicting laws
- government versus citizen's rights
- Gambling is allowed at some places whereas it is illegal at other places?

Network Hardware

- No generally accepted taxonomy into which all computer networks fit, but two dimensions stand out as important: **transmission technology** and **scale**.
- Transmission technology that are in widespread use:

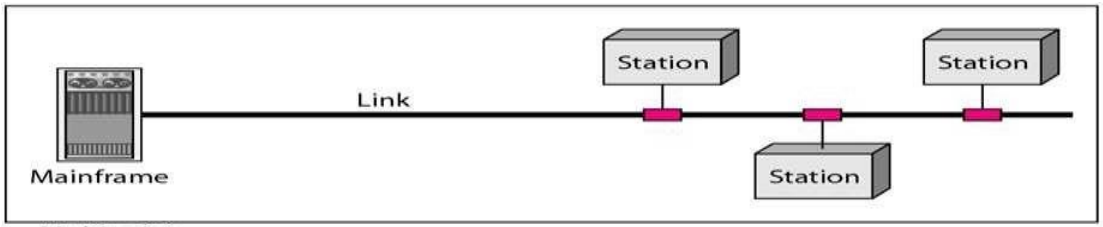
  Point-to-point links & Broadcast links
- **Point-to-point links**
- Point-to-point links connect individual pairs of machines.
- To go from the source to the destination on a network made up of point-to-point links, short messages, called **packets**, may have to first visit one or more intermediate machines.
- Ex. TV and remote control communication
- Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called **unicasting.**



Point-to-point

Network Hardware

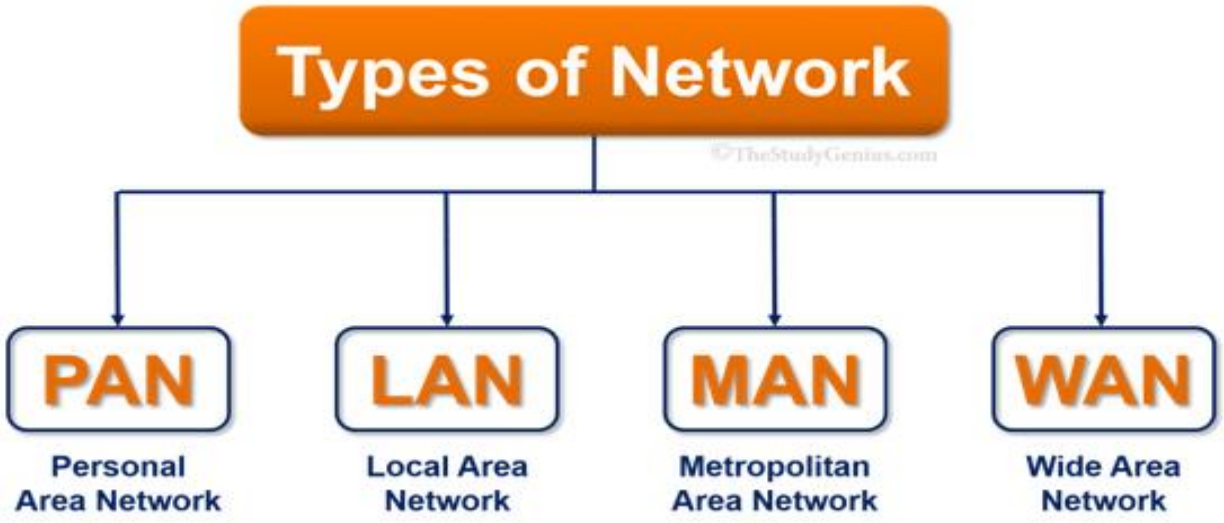
- **Broadcast**
- the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others.
- An address field within each packet specifies the intended recipient.
- Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored.
- A wireless network is a common example of a broadcast link, with communication shared over a coverage region.
- Some broadcast systems also support transmission to a subset of the machines (a specific group of stations), is known as **multicasting.** (This is a selective process)



Multipoint

Network Types

- Networks can be classified by their scale:
  - Vicinity PAN(Personal Area Network)
  - Building LAN (Local Area Network)
  - City MAN (Metropolitan Area Network)
  - Country WAN (Wide Area Network)
  - Planet The Internet (network of all networks)



Network Types

- Classifying networks is by **scale**.
- **Distance** is important as a classification metric because different technologies are used at different scales.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Figure 1-6. Classification of interconnected processors by scale.

Network Types

- **PAN (Personal Area Networks)**
- Communication over the range of person – very short distance
- Ex. Wireless connection between computer and peripherals – Bluetooth, Mobile phone –headset, car tape connection, RFID – on smart cards and library books
- **LAN (Local Area Networks)**

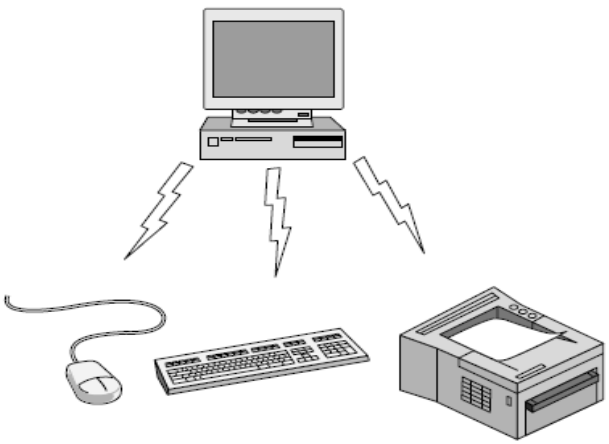


Figure 1-7. Bluetooth PAN configuration.

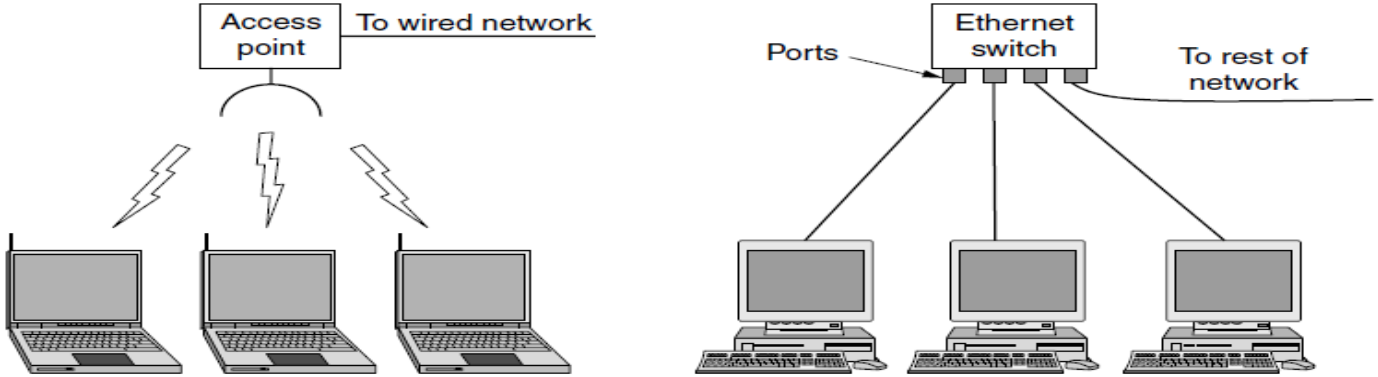


Figure 1-8. Wireless and wired LANs. (a) 802.11. (b) Switched Ethernet.

Network Types

- **LAN (Local Area Networks)**
- Privately owned network – operates within and nearby building like home, office, factory
- connect personal computers and consumer electronics to let them share resources (e.g., printers) and exchange information
- LANs used by companies - are called **enterprise networks**
- Wireless LAN – no cabling – popular
- Standard for wireless LANs - **IEEE 802.11,** popularly known as **WiFi**
- Wired LANs use copper wires, optical fiber
- wired LANs run at speeds of 100 Mbps to 1 Gbps, newer up to 10 Gbps
- Wired LAN performance better over wireless
- **IEEE 802.3**, popularly called **Ethernet,** is the most common type of wired LAN
- possible to divide one large physical LAN into two smaller logical LANs.
- Ex. Company- Engg. & Finance dept – Virtual LAN (VLAN) separate for each

Network Types

MAN (Metropolitan Area Networks)

- covers a city
- Example - cable television networks available in many cities
- wireless Internet access have resulted in another MAN, which has been standardized as **IEEE 802.16** and is popularly known as **WiMAX**

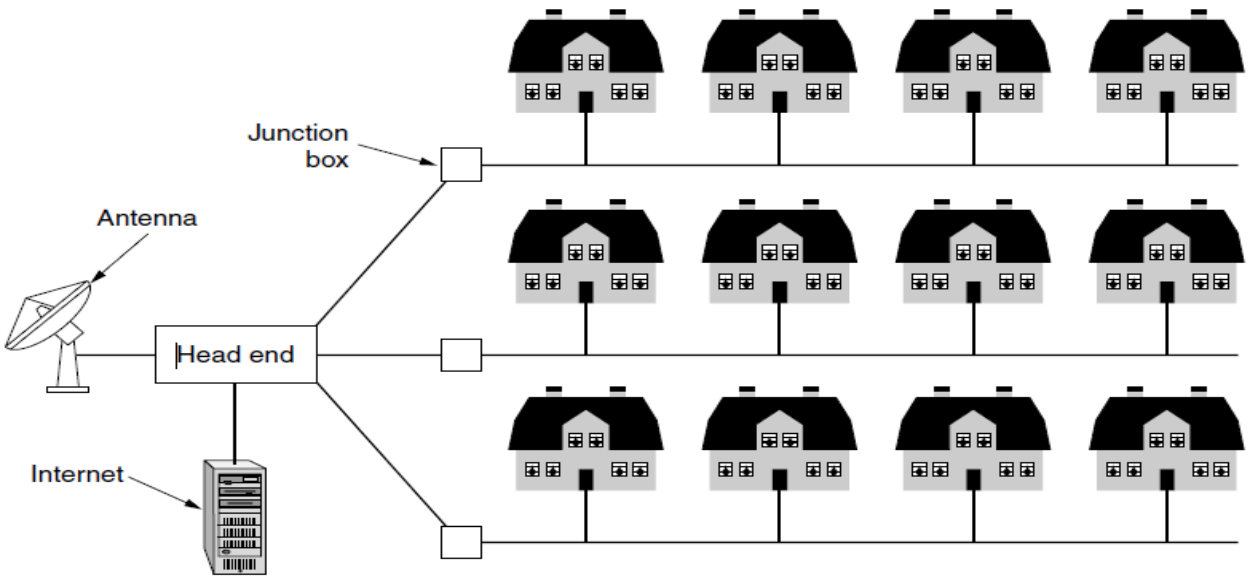


Figure 1-9. A metropolitan area network based on cable TV.

Activate
Go to PC se

Network Types

- **WAN (Wide Area Network)**
- Spans a large geographical area, often a country or continent.
- wired WANs example of a company with branch offices in different cities

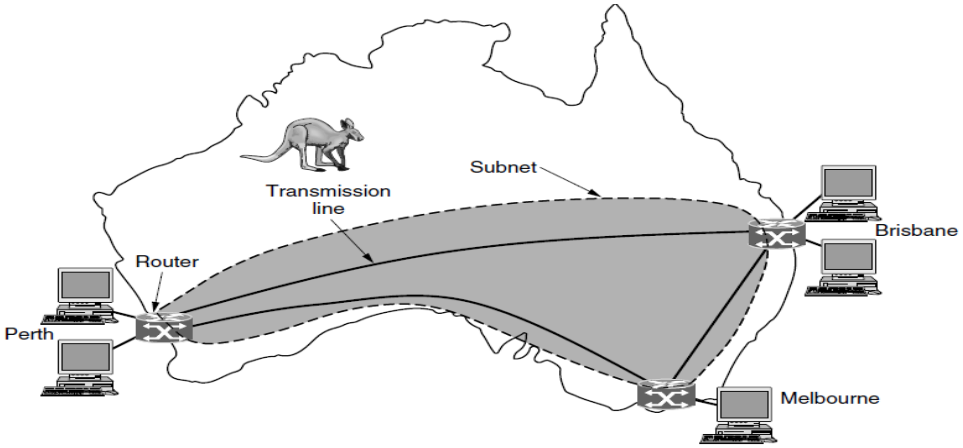


Figure 1-10. WAN that connects three branch offices in Australia.

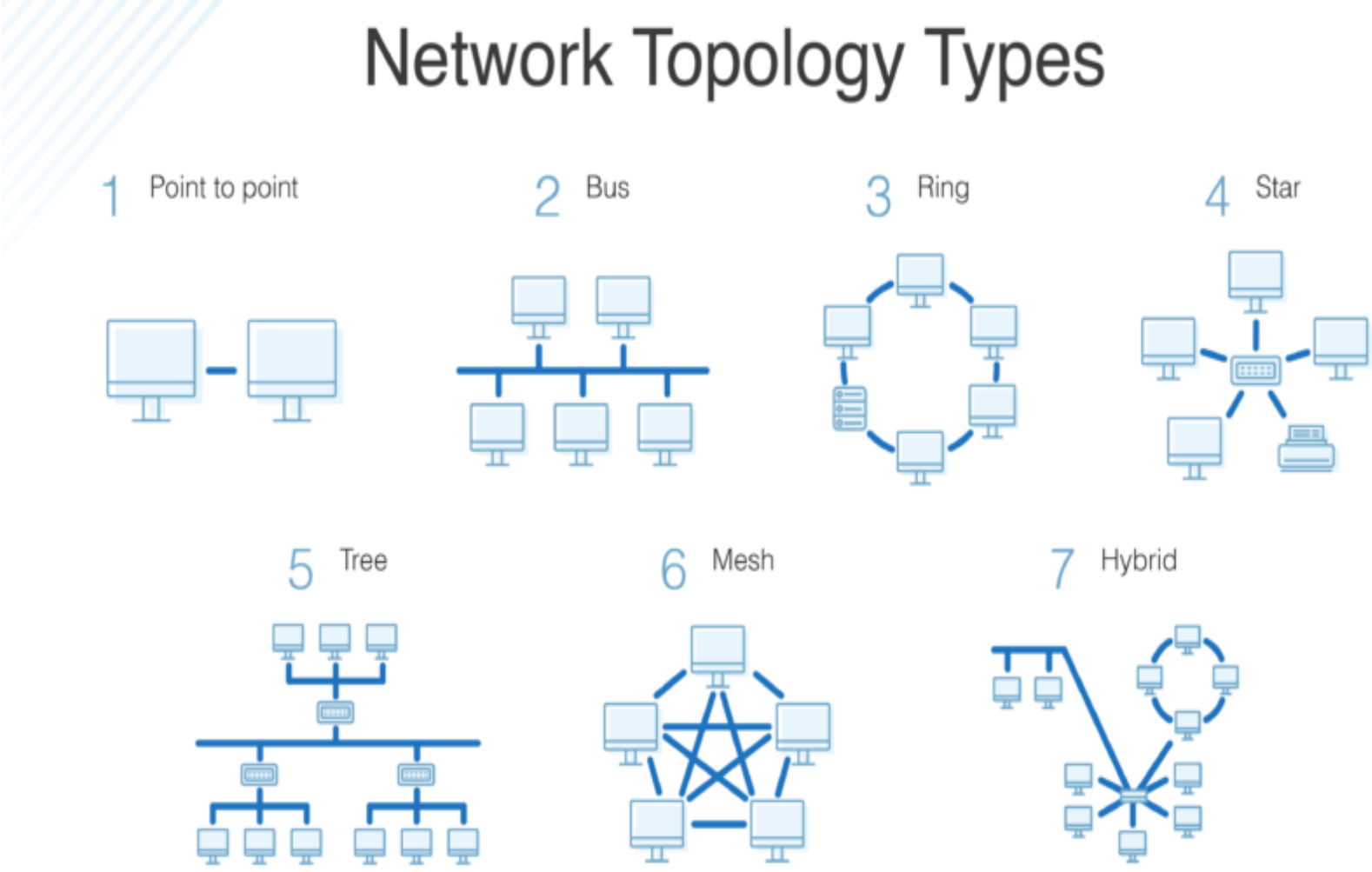
Activate

- **Internetworks**
- Many networks exist in the world, often with different hardware and software.
- People connected to one network often want to communicate with people attached to a different one.
- A collection of interconnected networks is called an **internetwork or internet**

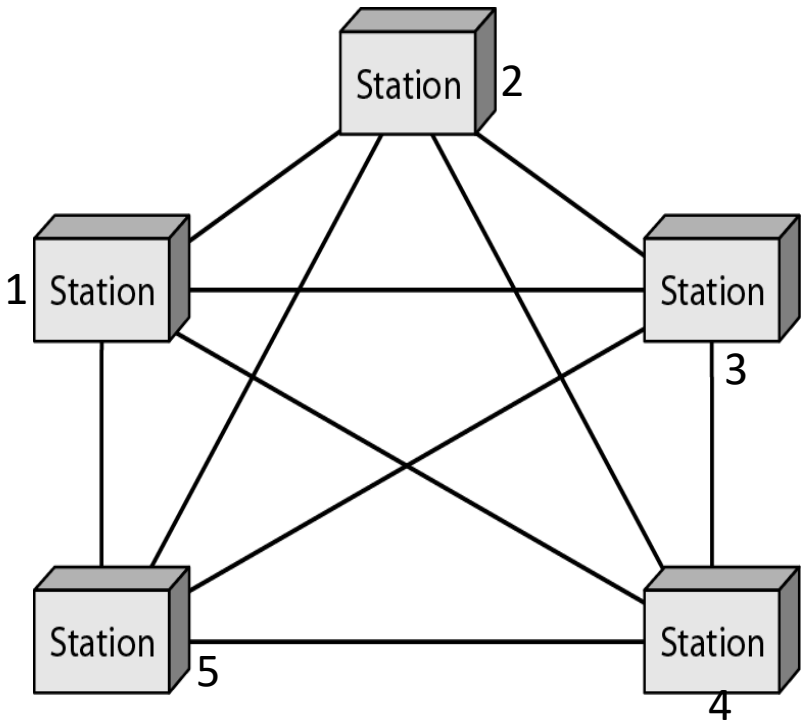
Network Topology

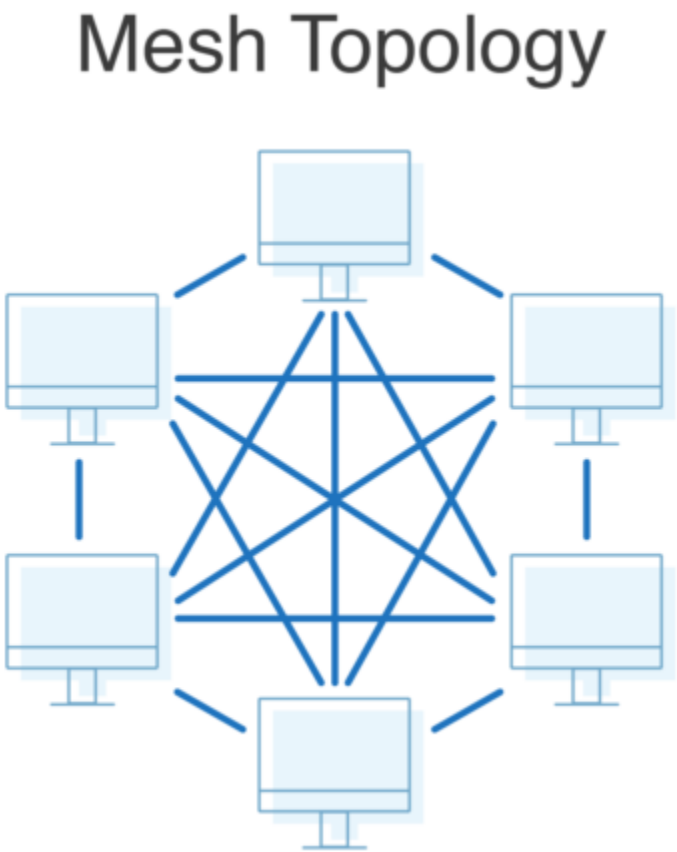
- “What is network topology?” - two categories in the network topology.
- **Physical** – The physical network topology refers to the actual connections (wires, cables, etc.) of how the network is arranged. Setup, maintenance, and provisioning tasks require insight into the physical network.
- **Logical** – The logical network topology is a higher-level *idea* of how the network is set up, including which nodes connect to each other and in which ways, as well as how data is transmitted through the network. Logical network topology includes any virtual and cloud resources.

Network Topology



Network Topology - Mesh





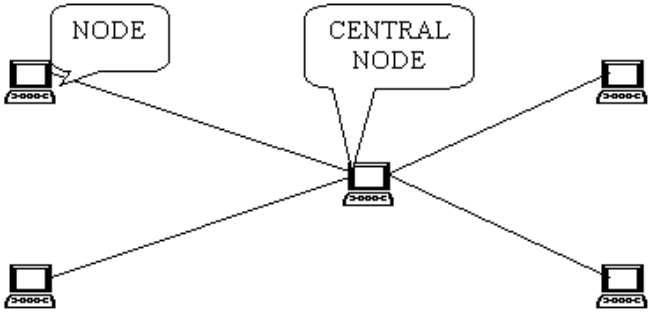
Pro: robustness

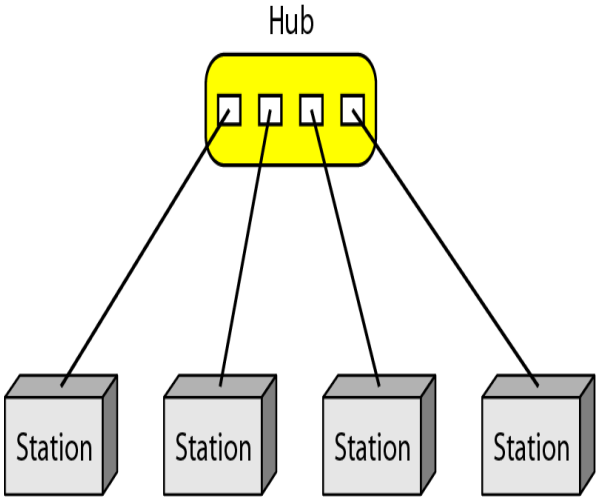
Con: many cabling/installation and need many I/O ports on nodes

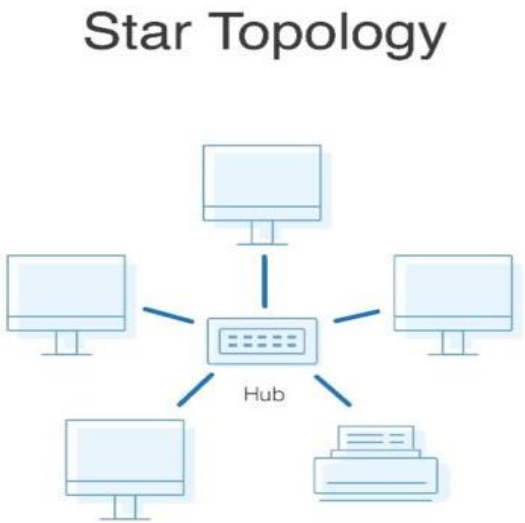
Network Topology - Mesh

- A mesh topology is an intricate and elaborate structure of point-to-point connections where the nodes are interconnected.
- Mesh networks can be full or partial mesh. Partial mesh topologies are mostly interconnected, with a few nodes with only two or three connections, while full-mesh topologies are fully interconnected.
- **Advantages of Mesh Topology**
- Mesh topologies are reliable and stable, and the complex degree of interconnectivity between nodes makes the network resistant to failure. For instance, no single device going down can bring the network offline.
- **Disadvantages of Mesh Topology**
- Mesh topologies are incredibly labor-intensive. Each interconnection between nodes requires a cable and configuration once deployed, so it can also be time-consuming to set up.
- Cost of cabling

Network Topology - Star







Example: Ethernet switch/hub, Wifi access point

Pro: robust against a node's failure or a link failure

Con: single point of failure at the hub

Network Topology - Star

- In star topology a number of workstations are directly linked to a central node
- Any communication between stations on a star LAN must pass through the central node.
- The central node controls all the activities of the nodes.

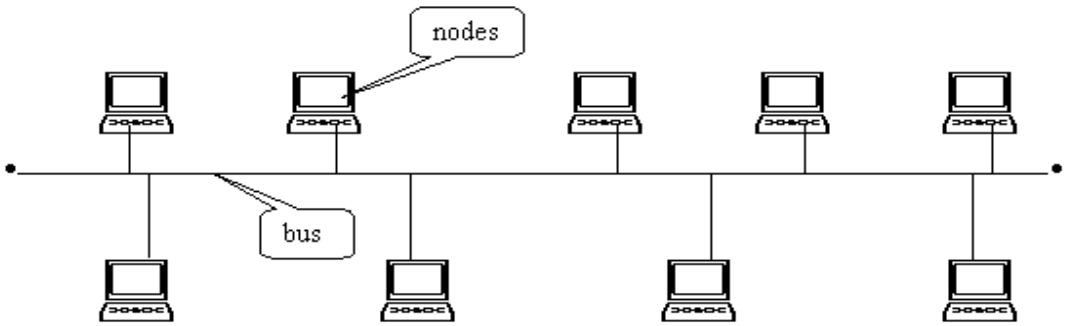
The advantages of the star topology are:

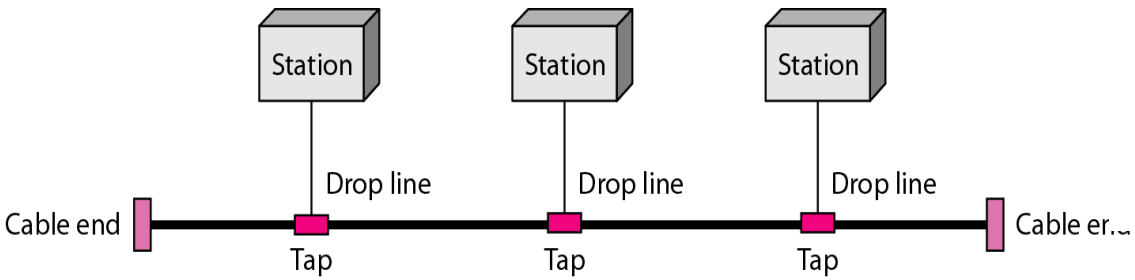
- It offers flexibility of adding or deleting of workstations from the network.
- Breakdown of one station does not affect entire system

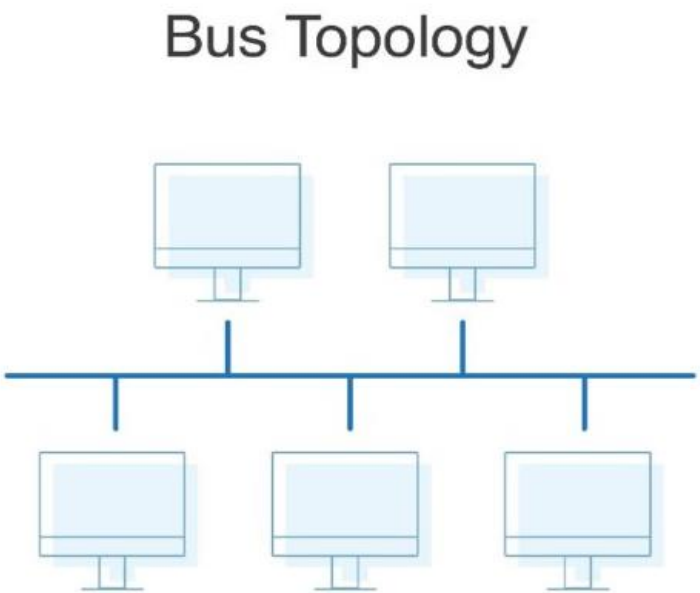
The major disadvantage of star topology is that failure of the central node disables communication throughout the whole network

The overall bandwidth and performance of the network are also limited by the central node's configurations and technical specifications, making star topologies expensive to set up and operate.

Network Topology - Bus







Example: first generation Ethernet that using cable for wiring

Con: any tap device failure will cause the whole system fail

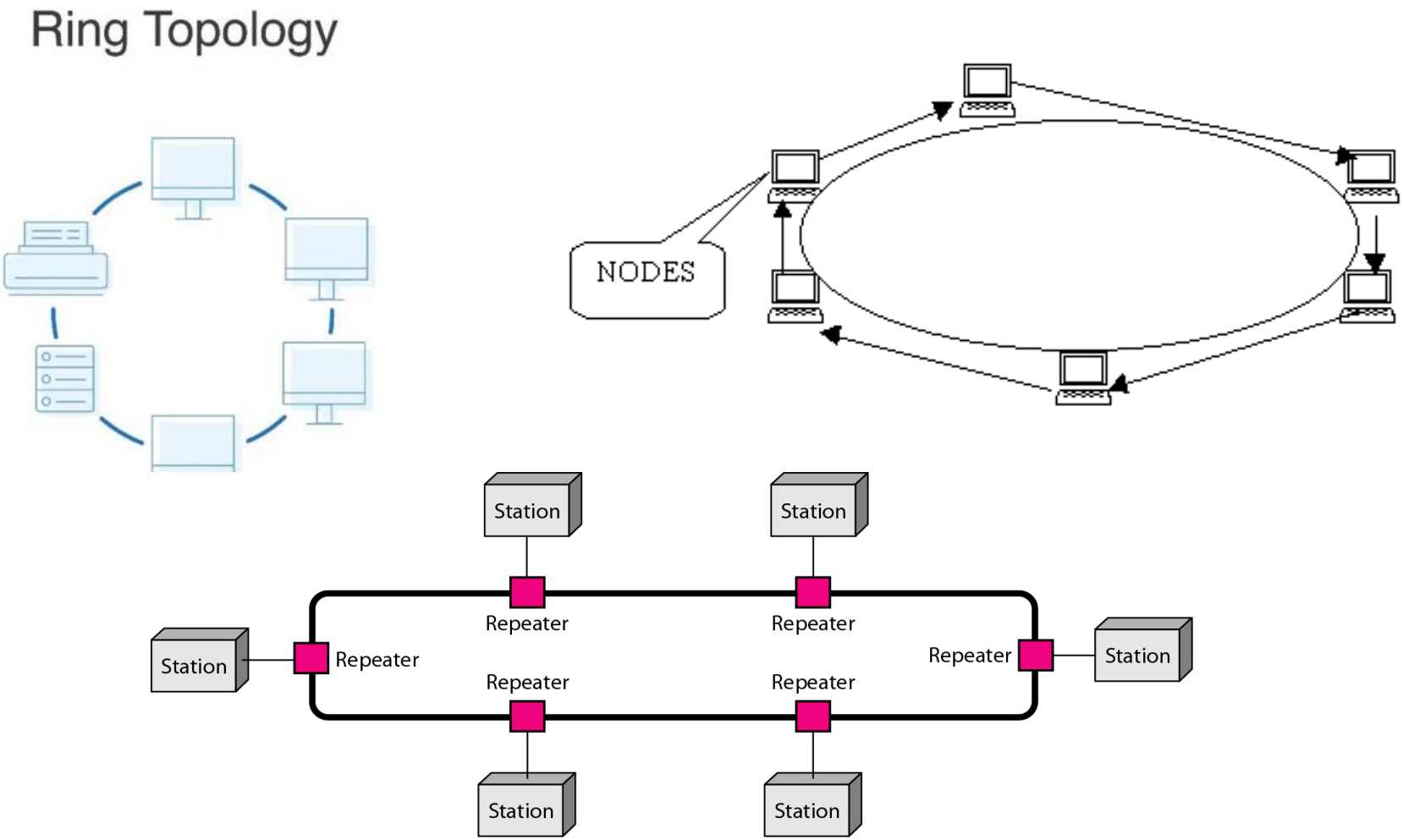
Network Topology - Bus

- In bus topology all workstations are connected to a single communication line called *bus*.
- There is no central node as in star topology.
- Transmission from any station travels the length of the bus in both directions and can be received by all workstations.
- Bus topologies are best suited for small networks because there's only so much bandwidth, and every additional node will slow transmission speeds.

The advantage of the bus topology is that

- It is quite easy to set up.
- If one station of the topology fails it does not affect the entire system.
- The **disadvantage of bus topology** is that any break in the bus is difficult to identify.

Network Topology - Ring



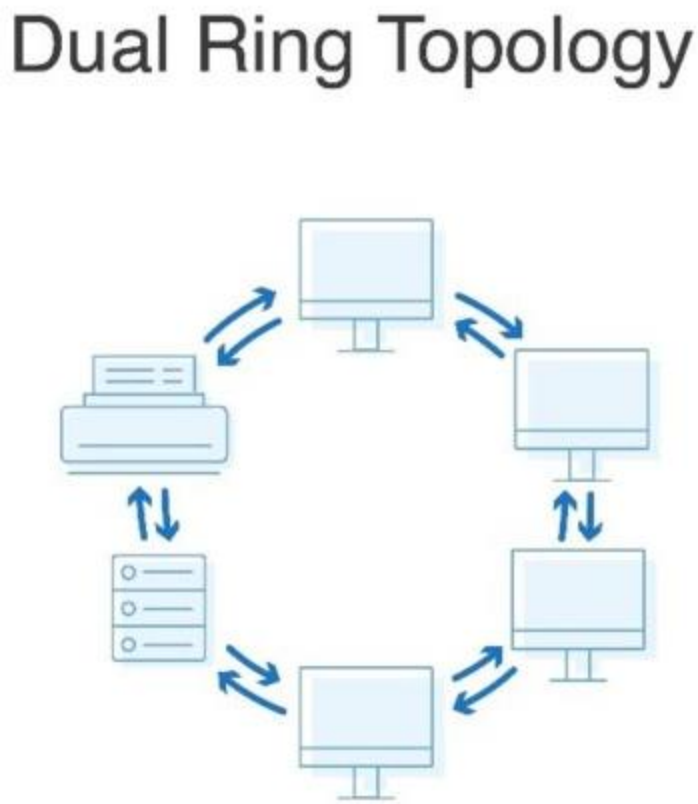
Network Topology - Ring

- In ring topology each station is attached nearby stations on a point to point basis so that the entire system is in the form of a ring.
- In this topology data is transmitted in one direction only.
- Thus the data packets circulate along the ring in either clockwise or anti-clockwise direction.

The advantage of this topology is that any signal transmitted on the network passes through all the LAN stations.

The disadvantage of ring network is that the breakdown of any one station on the ring can disable the entire system.

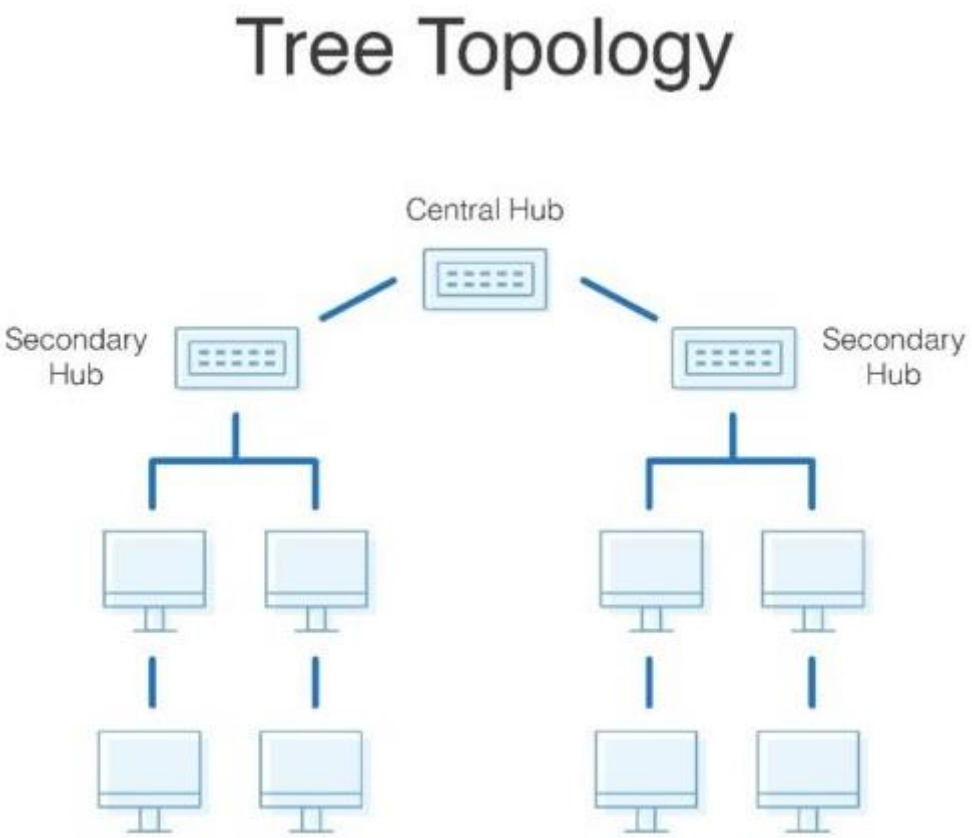
Network Topology – Dual Ring



Network Topology – Dual Ring

- **What Is Dual-Ring Topology?**
- A network with ring topology is half-duplex, meaning data can only move in one direction at a time. Ring topologies can be made full-duplex by adding a second connection between network nodes, creating a dual ring topology.
- **Advantages of Dual-Ring Topology**
- Efficiency: because each node has two connections on either side, information can be sent both clockwise and counterclockwise along the network.
- The secondary ring included in a dual-ring topology setup can act as a redundant layer and backup, which helps solve for many of the disadvantages of traditional ring topology.
- Dual ring topologies offer a little extra security, too: if one ring fails within a node, the other ring is still able to send data.

Network Topology - Tree



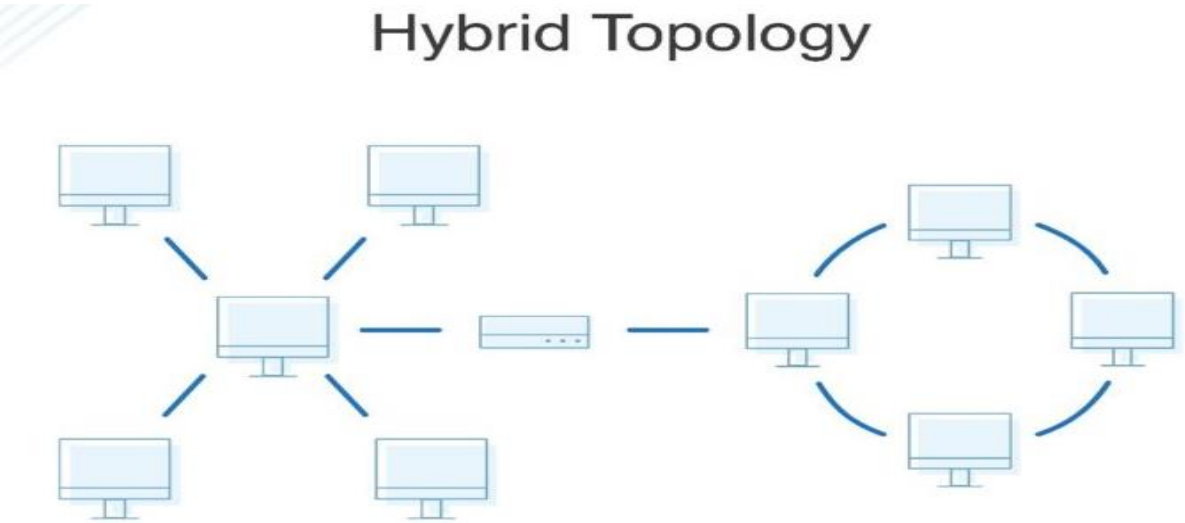
Network Topology – Tree

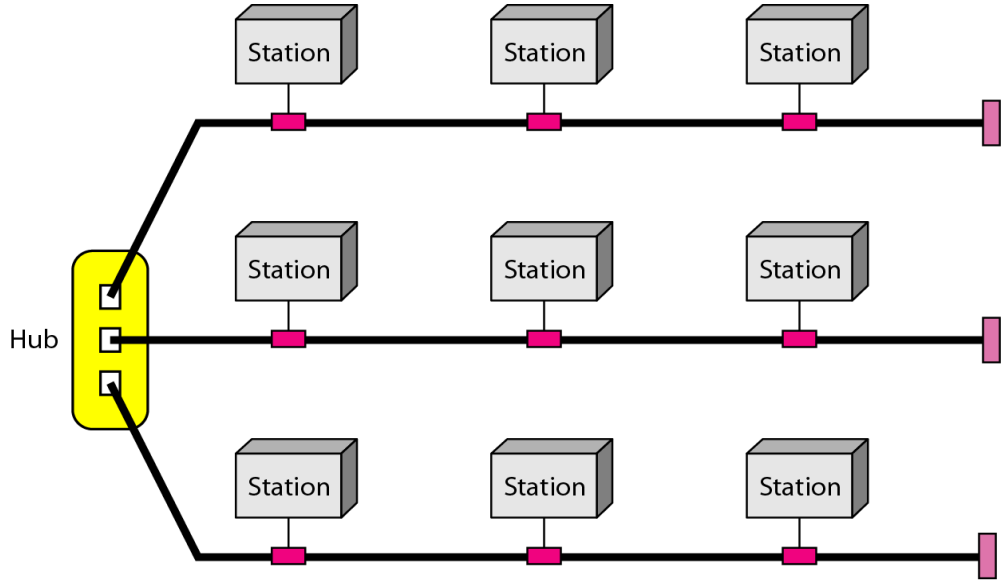
- **What Is Tree Topology?**
- The tree topology structure gets its name from how the central node functions as a sort of trunk for the network, with nodes extending outward in a branch-like fashion. However, where each node in a star topology is directly connected to the central hub, a tree topology has a parent-child hierarchy to how the nodes are connected. Those connected to the central hub are connected linearly to other nodes, so two connected nodes only share one mutual connection. Because the tree topology structure is both extremely flexible and scalable, it’s often used for wide area networks to support many spread-out devices.

Network Topology – Tree

- **Pros of Tree Topology**
- Combining elements of the star and bus topologies allows for the easy addition of nodes and network expansion. Troubleshooting errors on the network is also a straightforward process, as each of the branches can be individually assessed for performance issues.
- **Cons of Tree Topology**
- As with the star topology, the entire network depends on the health of the root node in a tree topology structure. Should the central hub fail, the various node branches will become disconnected, though connectivity within—but not between—branch systems will remain.
- Tree topologies are expensive because of the sheer amount of cabling required to connect each device to the next within the hierarchical layout.

Network Topology – Hybrid





Network Topology – Hybrid

- **What Is Hybrid Topology?**
- Hybrid topologies combine two or more different topology structures—the tree topology is a good example, integrating the bus and star layouts. Hybrid structures are most commonly found in larger companies where individual departments have personalized network topologies adapted to suit their needs and network usage.
- **Advantages of Hybrid Topology**
- The main advantage of hybrid structures is the degree of flexibility they provide, as there are few limitations on the network structure itself that a hybrid setup can't accommodate.
- **Disadvantages of Hybrid Topology**
- However, each type of network topology comes with its own disadvantages, and as a network grows in complexity, so too does the experience and know-how required on the part of the admins to keep everything functioning optimally. There's also the monetary cost to consider when creating a hybrid network topology.

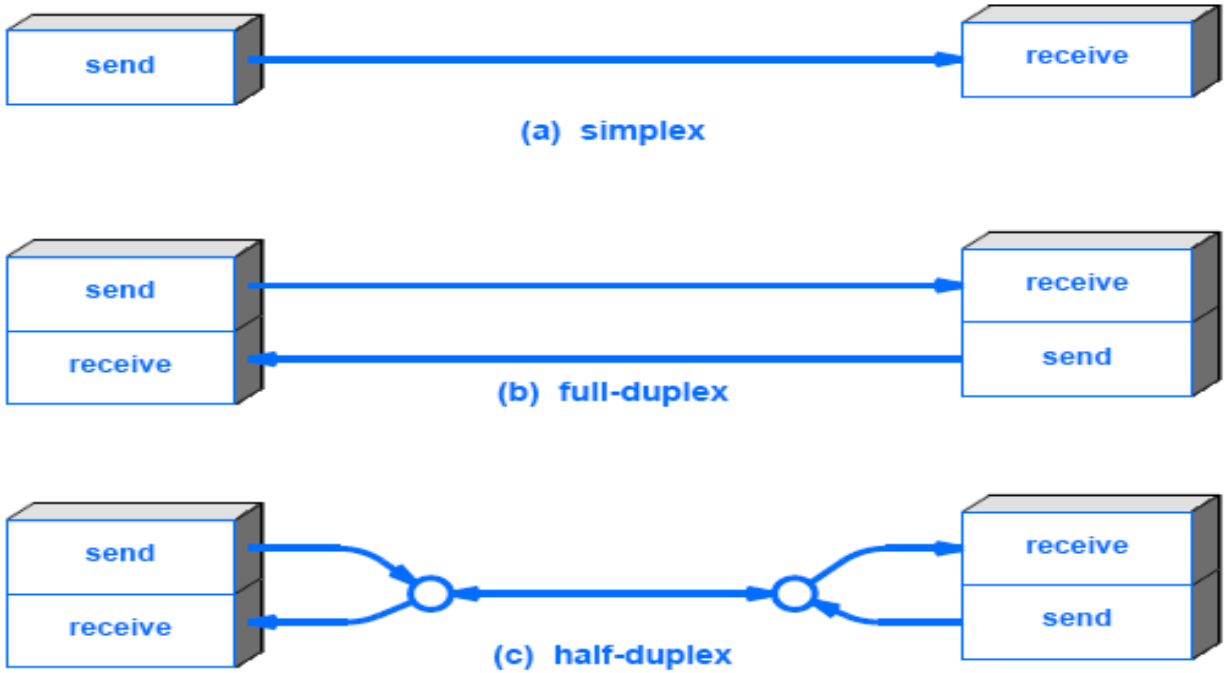
Data Flow (Simplex, Half Duplex, Full Duplex)

A communications channel can be classified as one of three types:

1. Simplex: A simplex mechanism can only transfer data in a single direction. It is analogous to broadcast radio or television

2. Full-Duplex: Full-duplex allows transmission in two directions simultaneously.

3. Half-Duplex: A half-duplex mechanism involves a shared trans-mission medium. The shared medium can be used for communication in each direction but the communication cannot proceed simultaneously.



Data Flow (Simplex, Half Duplex, Full Duplex)

- Simplex mode
  - Keyboard, mouse, traditional monitor
  - GPS device
- Half duplex
  - Walkie-talkie
  - Citizen band radio
  - Advantage: entire bandwidth can be used for transmission
  - Cons: not suitable for high interactive application
- Full duplex
  - Either two physically separate transmission paths or channel capacity is divided between signals travelling in both directions
  - Ex Telephone

NETWORK SOFTWARE

- Network software is an extremely broad term for a range of software aimed at the design, implementation and operation and monitoring of modern networks. Various types of network software support the creation, calibration and operation of networks
- **Functions of Network Software**
- Helps to set up and install computer networks
- Enables users to have access to network resources in a seamless manner
- Allows administrations to add or remove users from the network
- Helps to define locations of data storage and allows users to access that data
- Helps administrators and security system to protect the network from data breaches, unauthorized access and attacks on a network
- Enables network virtualizations

Protocol Hierarchies

- To reduce design complexity, most networks are organized as a stack of **layers** or **levels**
- The number of layers, name, contents and the function of each layer differ from network to network
- Each layer is a kind of virtual machine, offering certain services to the layer above it
- When layer *n on one machine carries on a conversation with layer n on another* machine, the rules and conventions used in this conversation are collectively known as the layer *n protocol*
- **Protocol** is an agreement between the communicating parties on how communication is to proceed

Protocol Hierarchies

- A five-layer network - Fig.
- The entities comprising the corresponding layers on different machines are called **peers.**
- The peers may be software processes, hardware devices, or even human beings
- No data are directly transferred from layer *n on one machine to* layer *n on another machine*
- Each layer passes data and control information to the layer immediately below it, until the lowest layer is reached
- Actual communication through physical medium

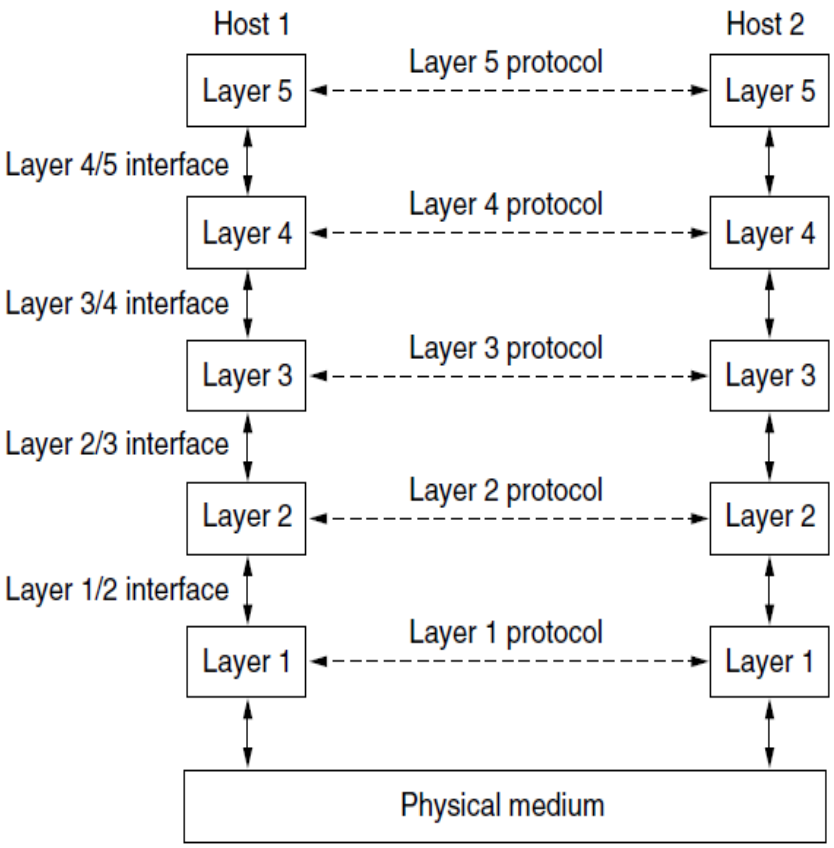


Figure 1-13. Layers, protocols, and interfaces.

Protocol Hierarchies

- Multilayer communication analogy

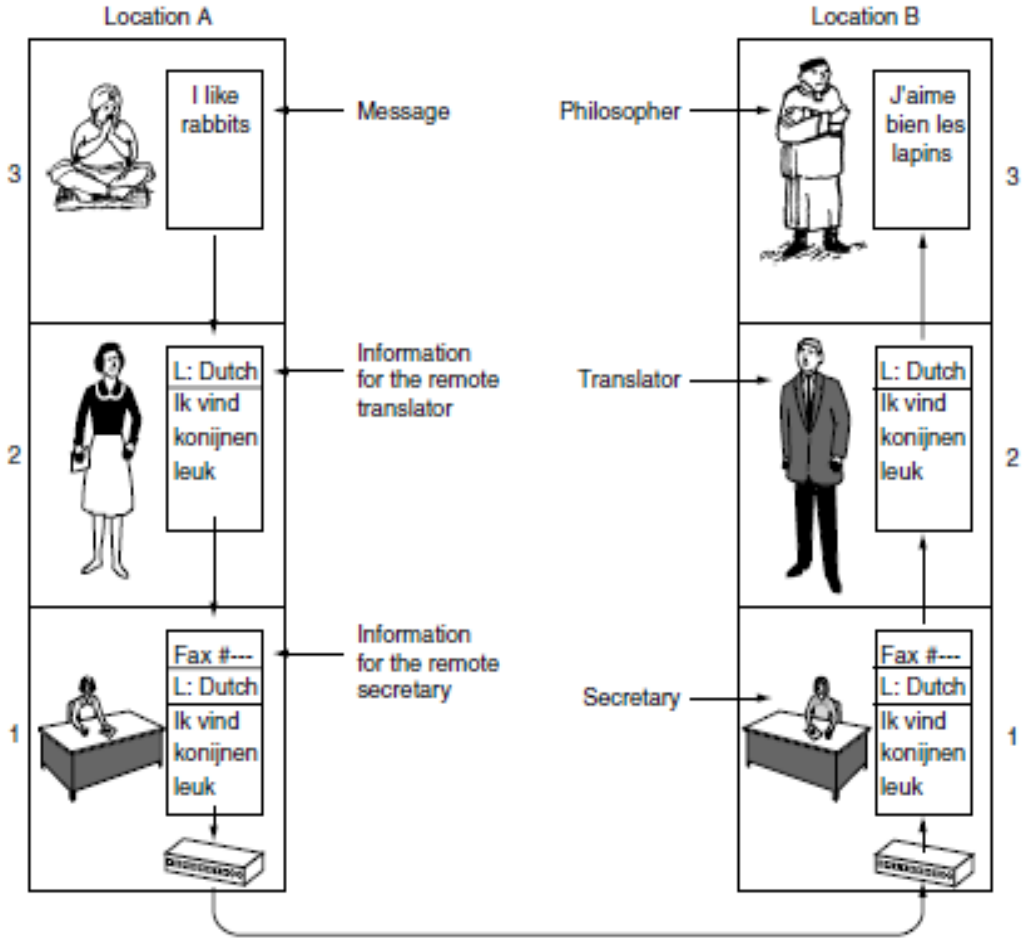


Figure 1-14. The philosopher-translator-secretary architecture.

Protocol Hierarchies

- Multilayer communication analogy
- Two philosophers (P) – (peer processes in layer 3)
  - P1 speaks Urdu and English and P2 speaks Chinese and French
- Translator (T) needed as no common language between two Ps - (peer processes at layer 2)
- T's contacts secretaries (S) (peer processes at layer 1)
- P1 wish to sends message " I like Rabbits" to P2, (peer to peer) so passes message in English to T1 (across interface 2/3)
- T1 and T2 agrees on a neutral language known to both – Dutch
- Choice of language is layer 2 protocol and is up to the layer 2 peer processes
- T1 gives message to S1 for transmission, S1 sends it to S2
- Its between S1 and S2, may communicate through email or so (layer 1 protocol)
- S2 forwards to T2, T2 translates in French and passes to P1

Protocol Hierarchies

- Communication to top layer of five-layer network

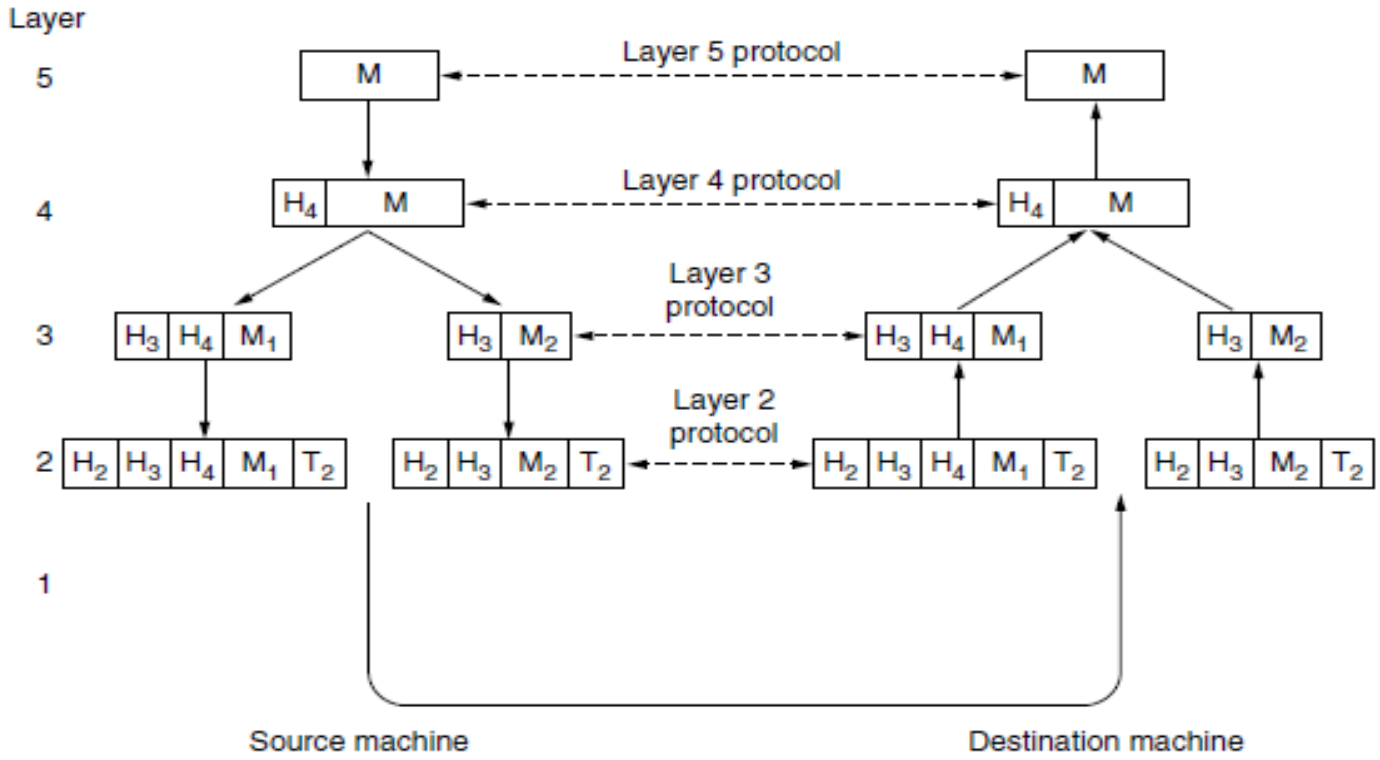


Figure 1-15. Example information flow supporting virtual communication in layer 5.

Protocol Hierarchies

- A message, *M, is produced by* an application process running in layer 5 and given to layer 4 for transmission.
- Layer 4 puts a **header** in front of the message to identify the message and passes the result to layer 3. The header includes control information, such as addresses, to allow layer 4 on the destination machine to deliver the message.
- Other examples of control information used in some layers are sequence numbers (in case the lower layer does not preserve message order), sizes, and times.
- layer 3 must break up the incoming messages into smaller units, packets, prepending a layer 3 header to each packet. In this example, *M is* split into two parts, *M1 and M2, that will be transmitted separately*
- Layer 3 decides which of the outgoing lines to use and passes the packets to layer 2.
- Layer 2 adds to each piece not only a header but also a trailer, and gives the resulting unit to layer 1 for physical transmission.
- At the receiving machine the message moves upward, from layer to layer, with headers being stripped off as it progresses. None of the headers for layers below *n are passed up to layer n.*

Design Issues for the Layers

- **Reliability**
  - **Error detection**
  - **Error correction**
  - **Routing**
- **Evolution of the Network**
  - **Addressing or naming**
  - **Scalability**
- **Resource allocation (sharing)**
  - **Multiplexing**
  - **Flow control**
  - **Congestion**
- **Timeliness delivery and quality of service**
- **Security**
  - **Confidentiality**
  - **Authentications**
  - **integrity**

Connection-Oriented Versus Connectionless Service

- Layers can offer two different types of service to the layers above them: **connection- oriented** and **connectionless**
- **connection-oriented** network service, the service user first establishes a connection, uses the connection, and then releases the connection.
- The essential aspect of a connection is that it acts like a tube: the sender pushes objects (bits) in at one end, and the receiver takes them out at the other end. In most cases the order is preserved so that the bits arrive in the order they were sent.
- Similar to telephone system (circuit)
- **connectionless** service is modeled after the postal system.
- Each message (letter) carries the full destination address and each one is routed through the intermediate nodes inside the system independent
- of all the subsequent messages. There are different names for messages in different contexts; a **packet** is a message at the network layer.

Connection-Oriented Versus Connectionless Service

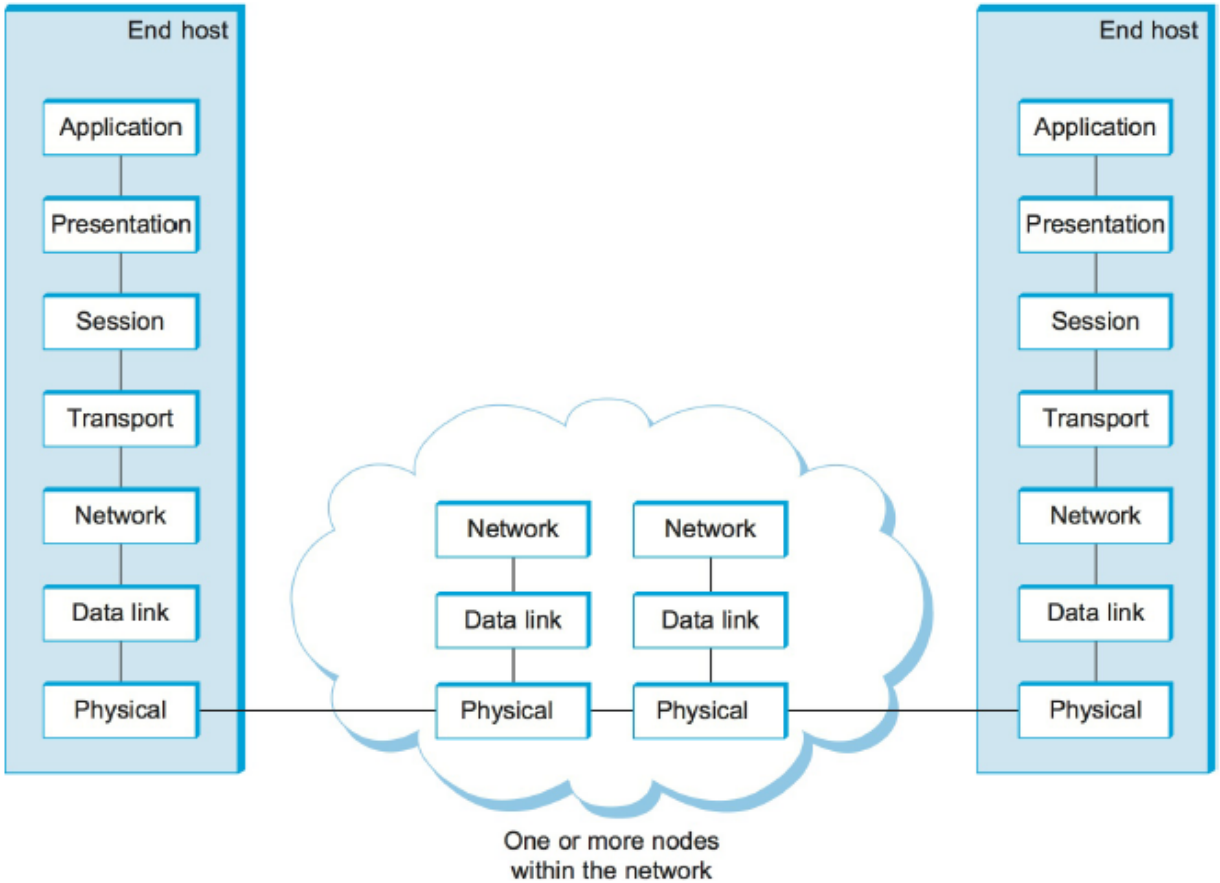
- When the intermediate nodes receive a message in full before sending it on to the next node, this is called **store-and-forward switching.**
- The alternative, in which the onward transmission of a message at a node starts before it is completely received by the node, is called **cut-through switching.**
- Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. However, it is possible that the first one sent can be delayed so that the second one arrives first.

Reference Models

- Two important network architectures:
- The OSI reference model and the TCP/IP reference model
- Although the *protocols associated* with the OSI model are not used any more, the *model itself is actually quite* general and still valid, and the features discussed at each layer are still very important.
- The TCP/IP model has the opposite properties: the model itself is not of much use but the protocols are widely used.

OSI- Reference Model

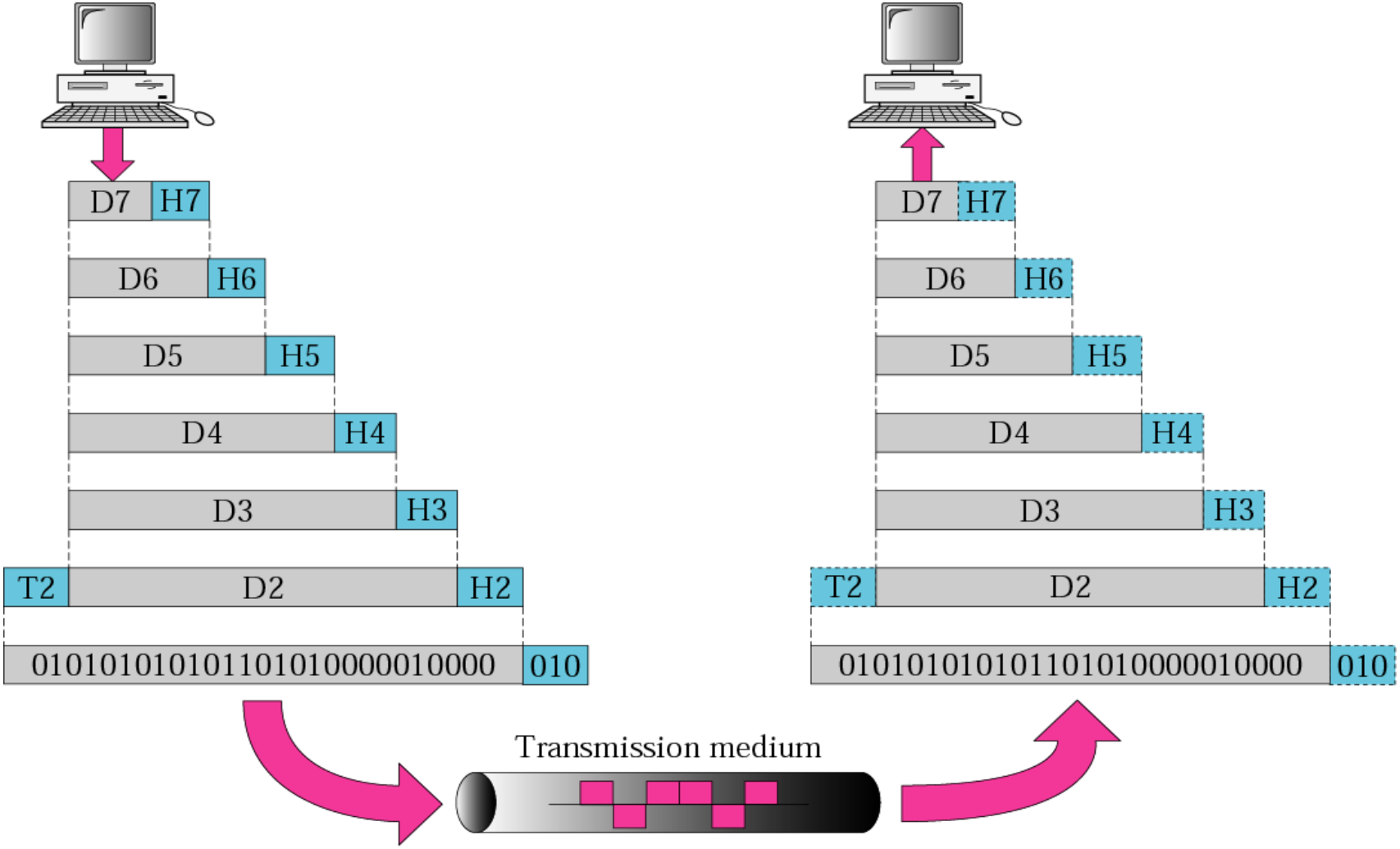
Layer 7	Application
Layer 6	Presentation
Layer 5	Session
Layer 4	Transport
Layer 3	Network
Layer 2	Data Link
Layer 1	Physical



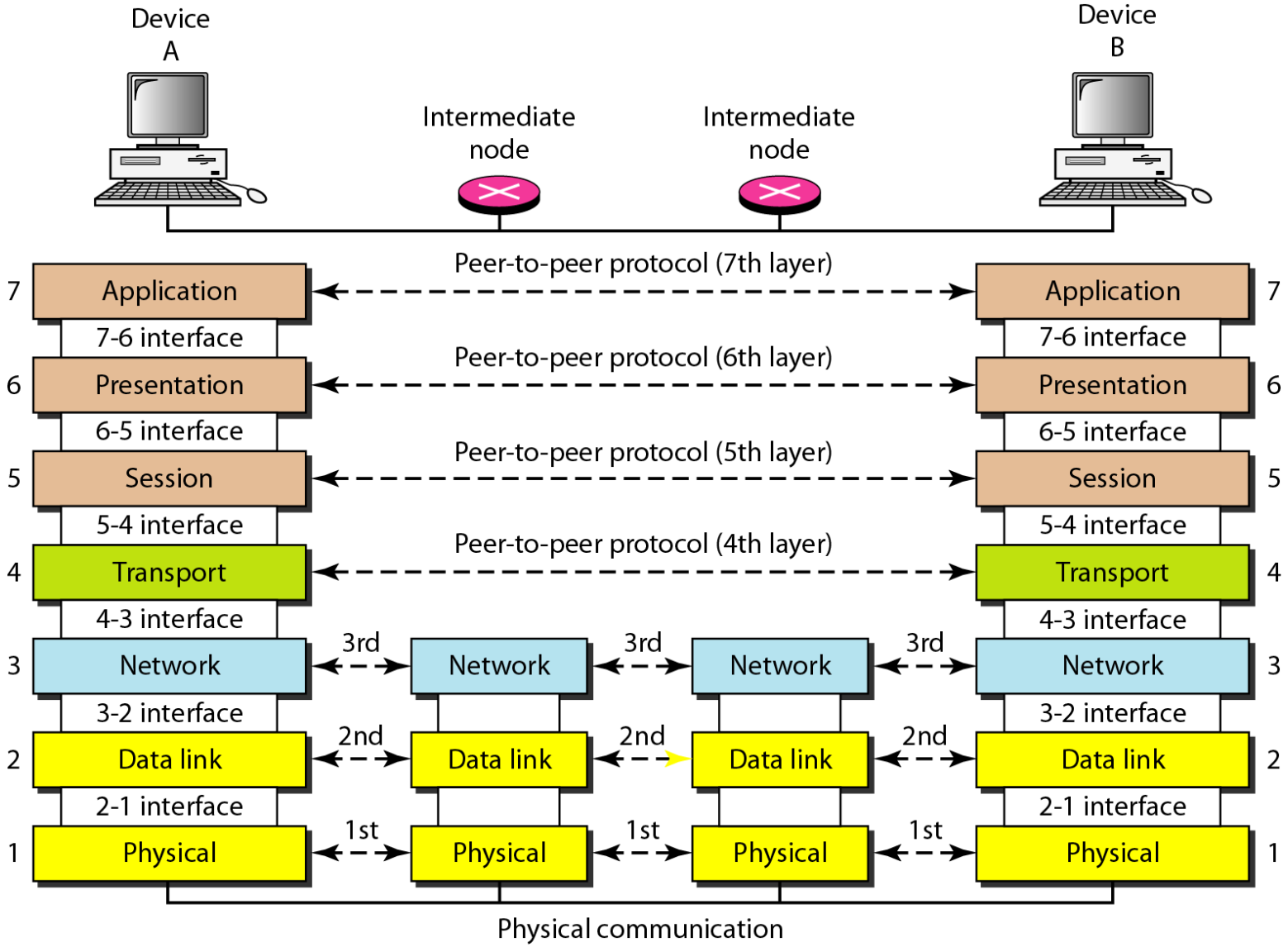
OSI Model

- **OSI** (**Open System Interconnection)** is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.
- **OSI** model was developed by the **I**nternational **O**rganization for **S**tandardization (**ISO**) in 1984, and it is now considered as an architectural model for the inter-computer communications
- The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software
- OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.
- Each layer is self-contained, so that task assigned to each layer can be performed independently
- ISO is the organization. OSI is the model

OSI Model


D7
H7
D6
H6
D5
H5
D4
H4
D3
H3
T2
D2
H2
0101010101011010100000010000
010
Transmission medium

OSI Model

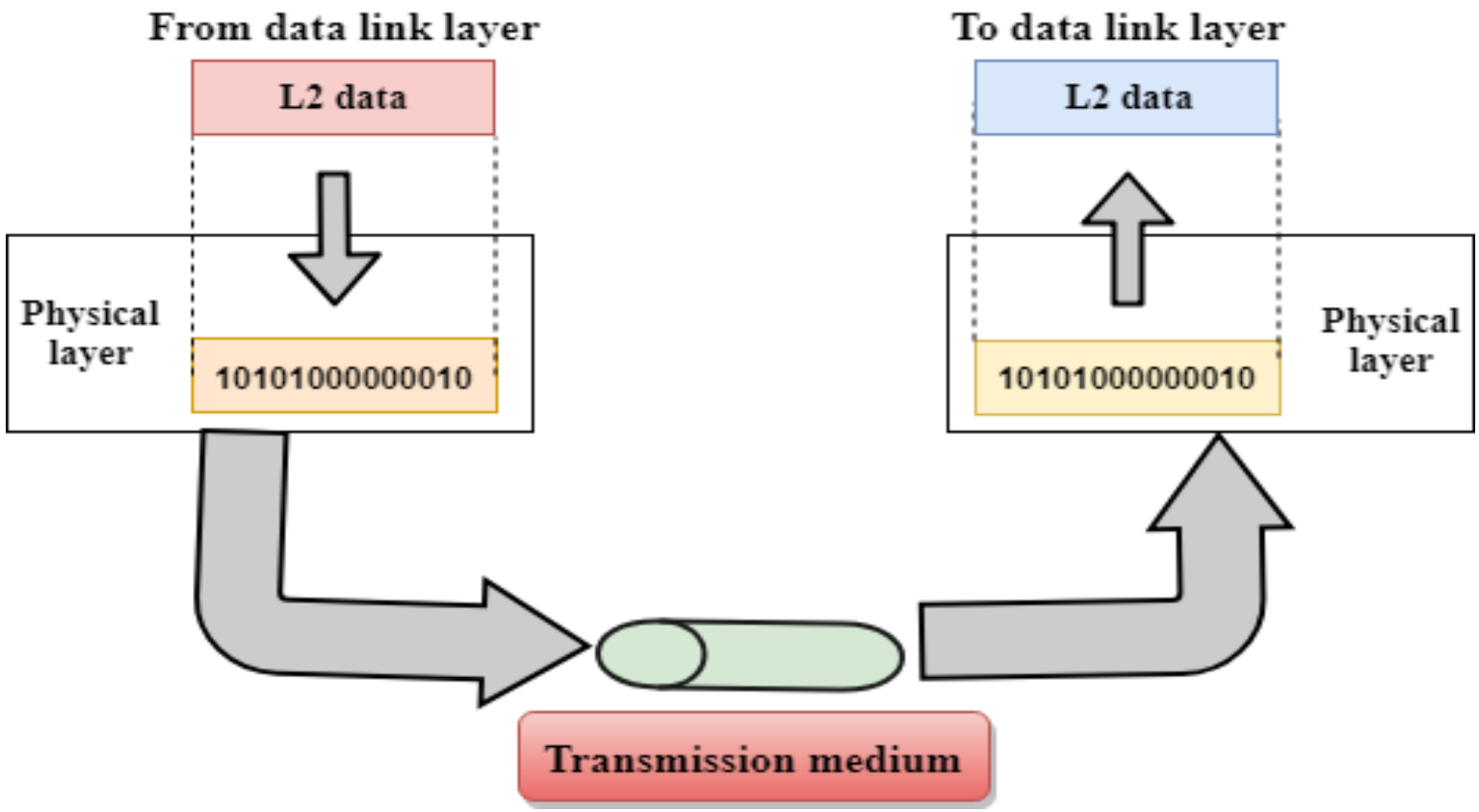

Device A
Device B
Intermediate node
Intermediate node
Peer-to-peer protocol (7th layer)
Peer-to-peer protocol (6th layer)
Peer-to-peer protocol (5th layer)
Peer-to-peer protocol (4th layer)
7 Application
7-6 interface
6 Presentation
6-5 interface
5 Session
5-4 interface
4 Transport
4-3 interface
3 Network
3-2 interface
2 Data link
2-1 interface
1 Physical
3rd
2nd
1st
Physical communication

ISO-OSI Reference Model

The OSI model has seven layers. The principles that were applied to arrive at the seven layers can be briefly summarized as :

- A layer should be created where a different abstraction is needed.
- Each layer should perform a well-defined function.
- The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
- The layer boundaries should be chosen to minimize the information flow across the interfaces.
- The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

Physical layer



Physical Layer

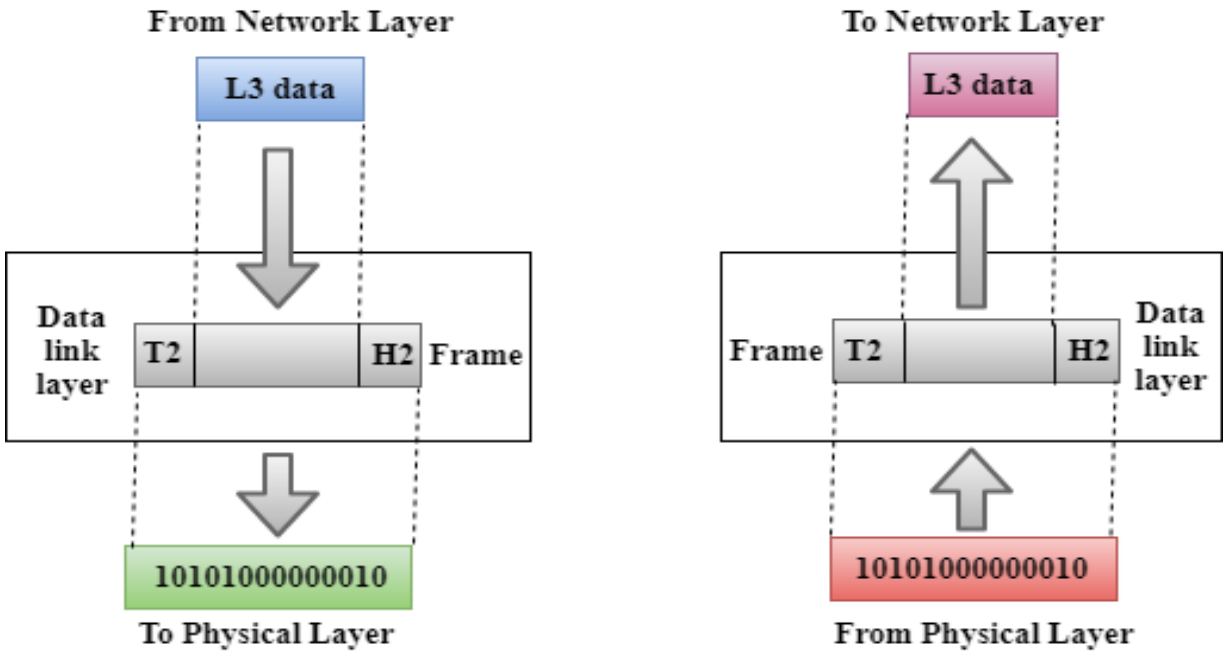
- Bottom layer
- The **physical layer** is concerned with transmitting raw bits over a communication channel.
- The design issues have to do with making sure that when one side sends a 1 bit it is received by the other side as a 1 bit, not as a 0 bit.
- Typical questions here are what electrical signals should be used to represent a 1 and a 0, how many nanoseconds a bit lasts, whether transmission may proceed simultaneously in both directions, how the initial connection is established, how it is torn down when both sides are finished, how many pins the network connector has, and what each pin is used for.
- These design issues largely deal with mechanical, electrical, and timing interfaces, as well as the physical transmission medium, which lies below the physical layer.

Physical layer

Functions of a Physical layer:

- **Line Configuration:** It defines the way how two or more devices can be connected physically.
- **Data Transmission:** It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- **Topology:** It defines the way how network devices are arranged.
- **Signals:** It determines the type of the signal used for transmitting the information

Data-Link Layer



- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.
- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a local network

Data-Link Layer

- It contains two sub-layers:
- **Logical Link Control Layer**
  - It is responsible for transferring the packets to the Network layer of the receiver that is receiving.
  - It identifies the address of the network layer protocol from the header.
  - It also provides flow control.
- **Media Access Control Layer**
  - A Media access control layer is a link between the Logical Link Control layer and the network's physical layer.
  - It is used for transferring the packets over the network.

Data-Link Layer

Functions of the Data-link layer

- **Framing:** The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame. The header which is added to the frame contains the hardware destination and source address.

- **Physical Addressing:** The Data link layer adds a header to the frame that contains a destination address. The frame is transmitted to the destination address mentioned in the header.
- **Flow Control:** Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.

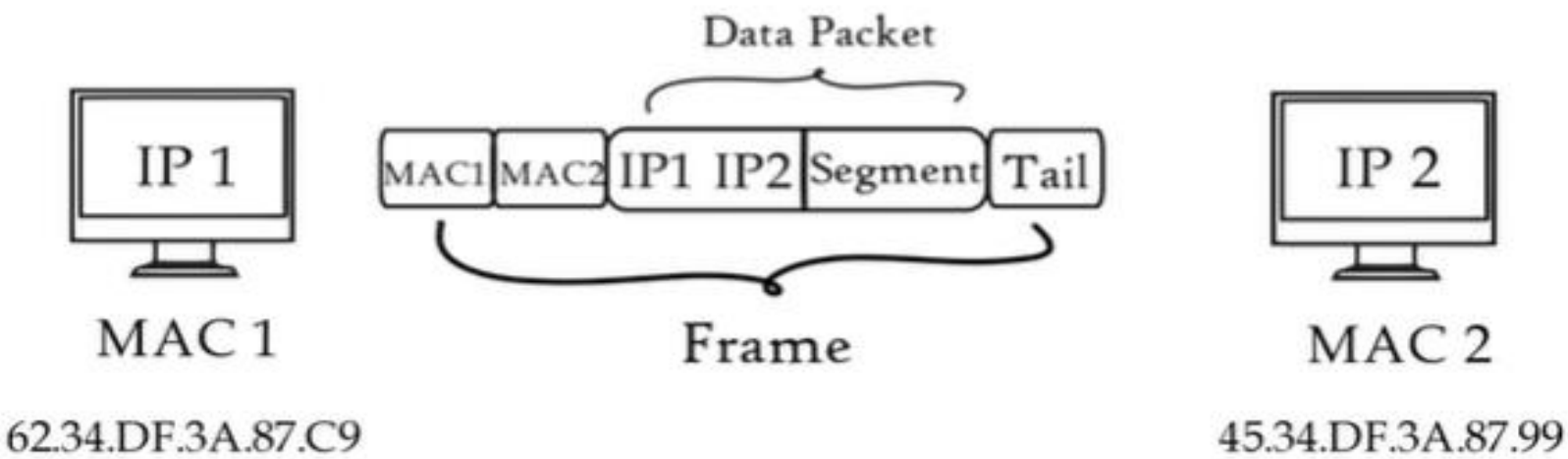
Data-Link Layer

- **Error Control:** Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer. If any error seems to occurr, then the receiver sends the acknowledgment for the retransmission of the corrupted frames.
- **Access Control:** When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

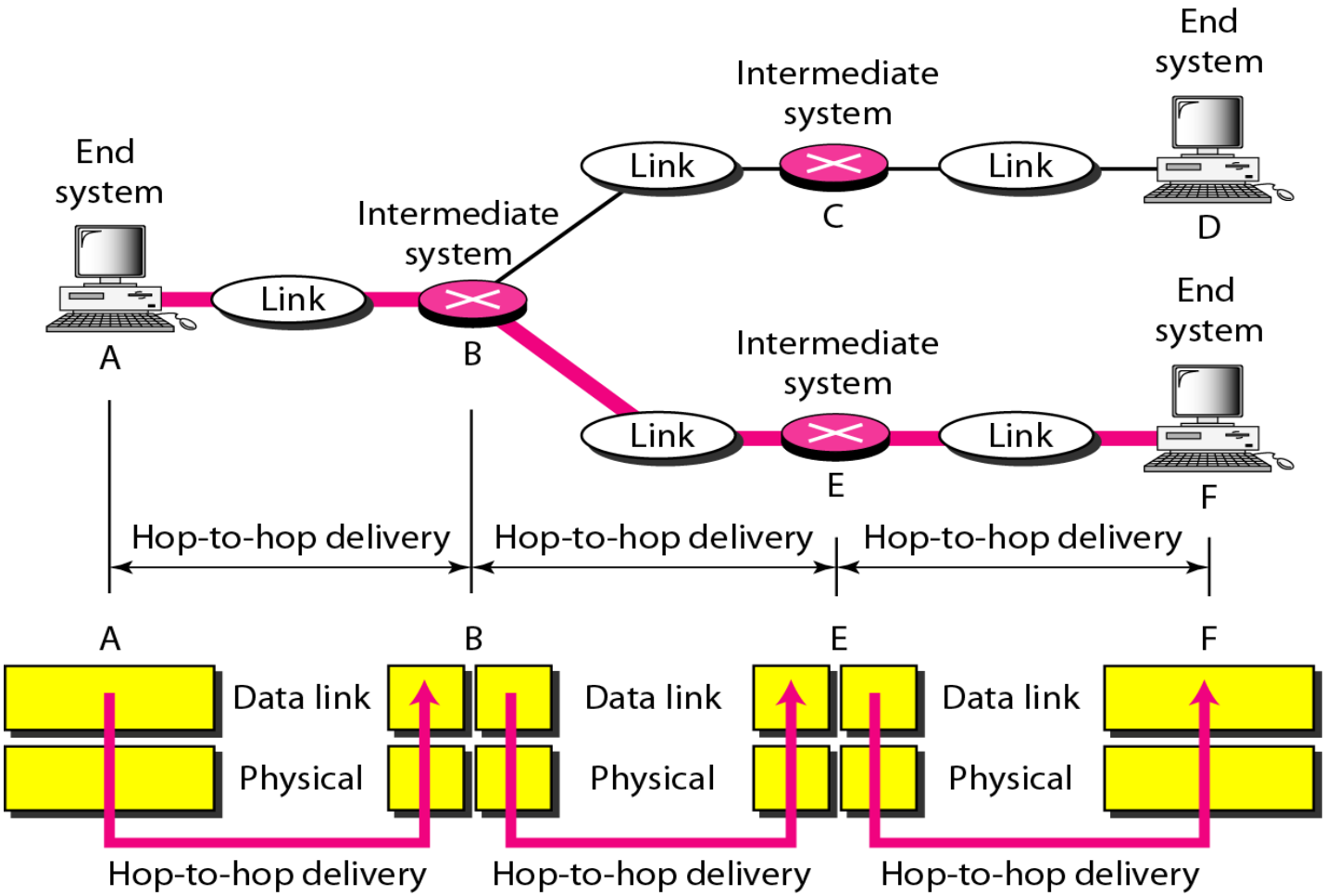
Data-Link Layer

Data Link Layer

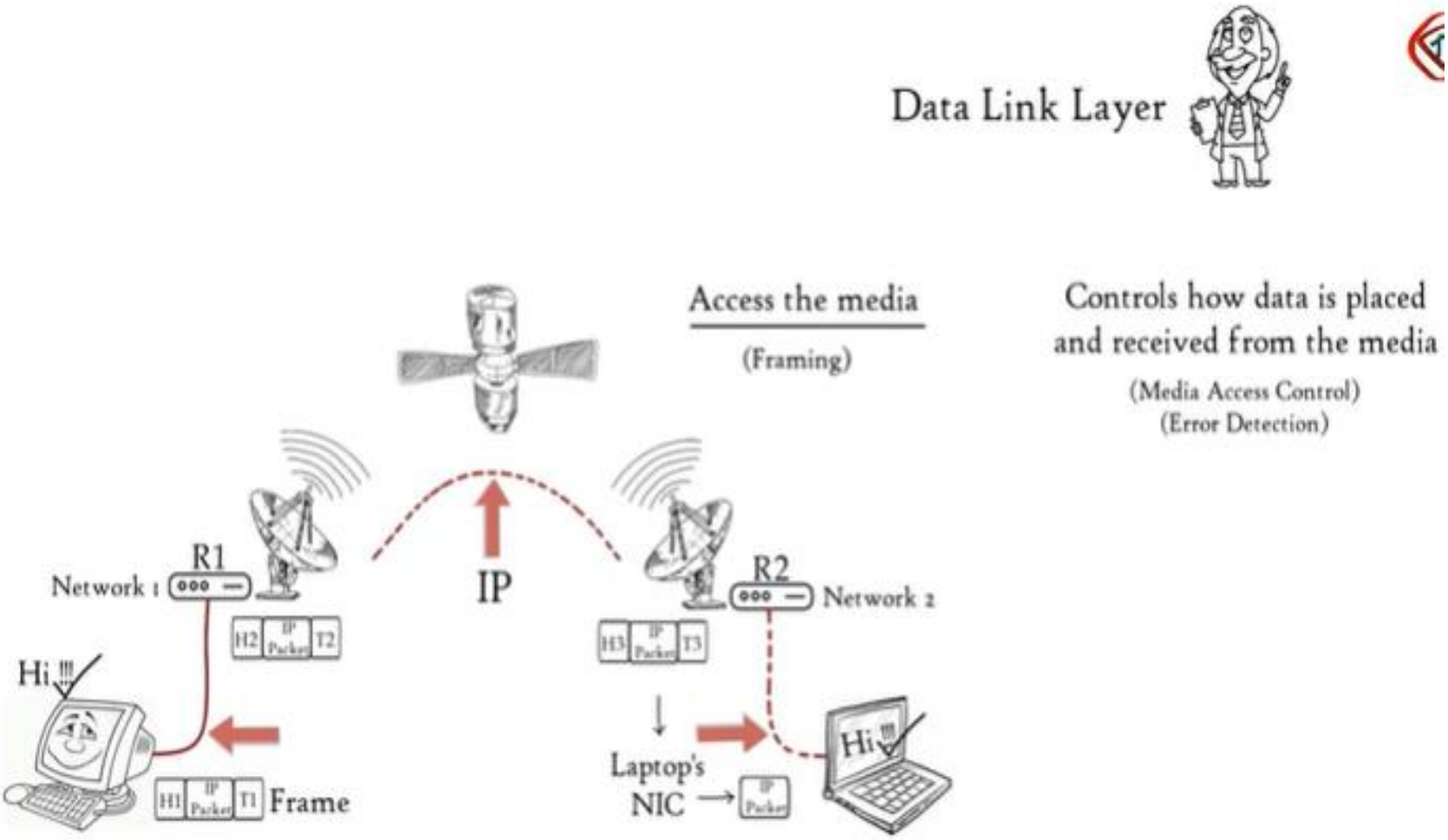
- Logical addressing : Network layer
- Physical addressing : Data Link layer



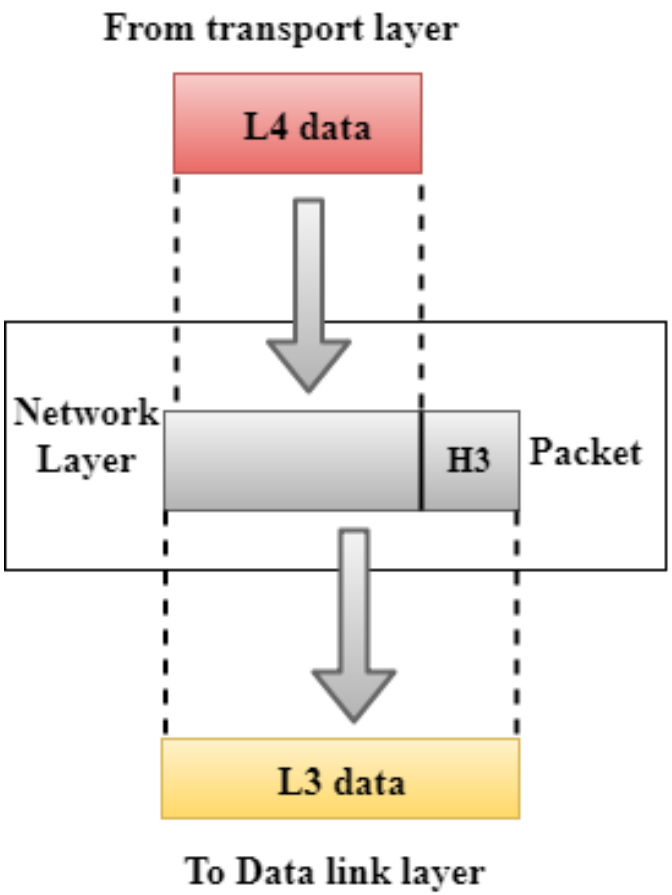
- Hop-to-hop delivery

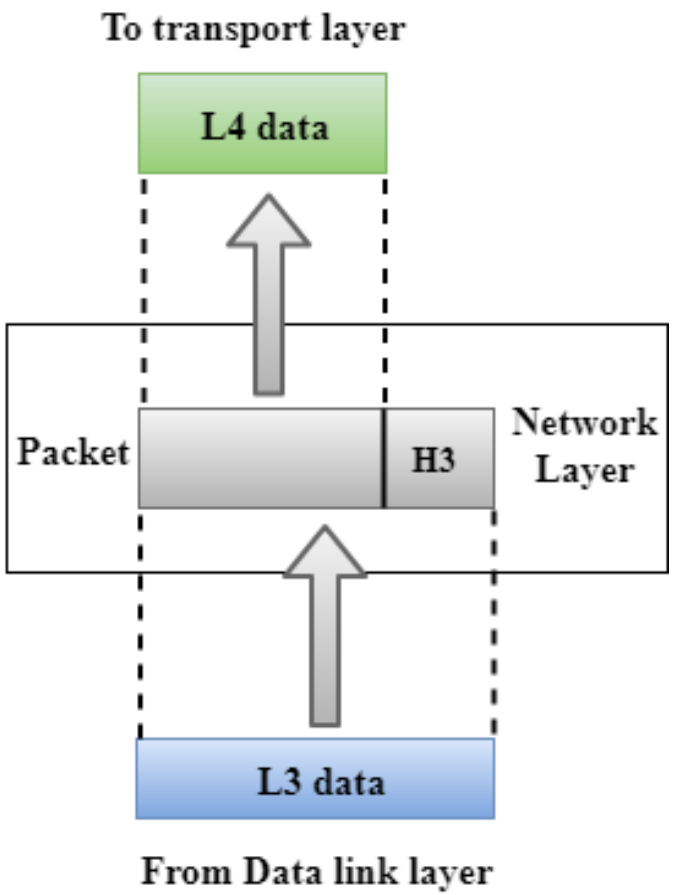


Data-Link Layer



Network Layer





Network Layer

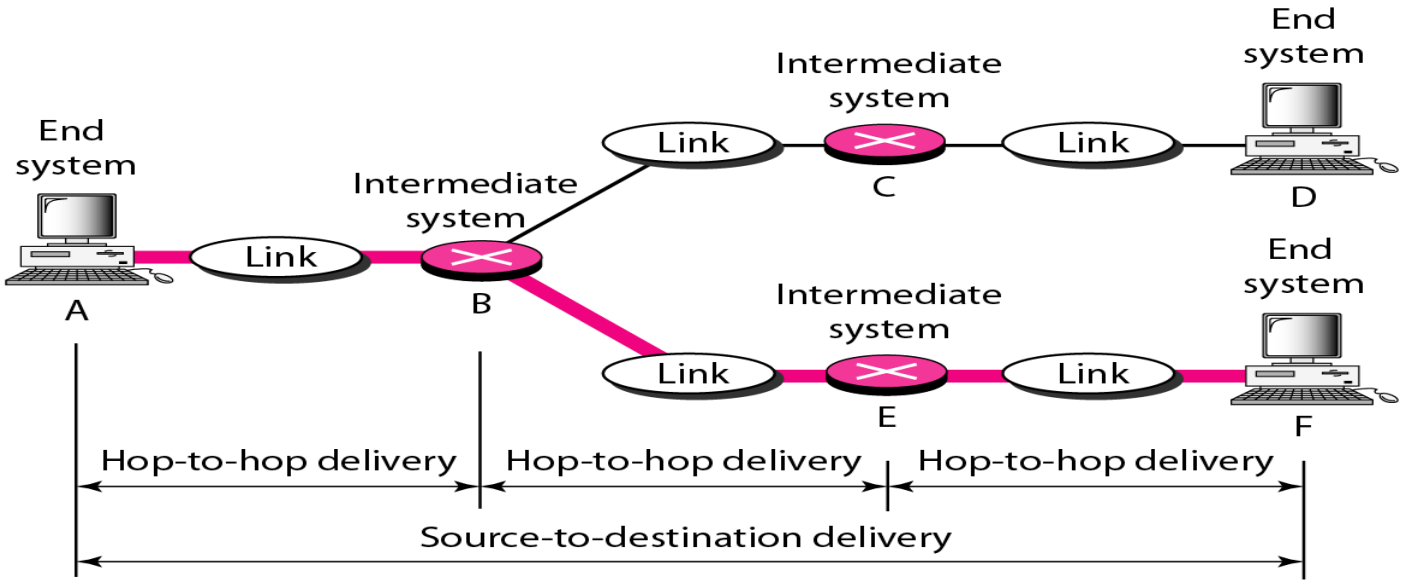
- It is a layer 3 that manages device addressing, tracks the location of devices on the network.
- A key design issue is determining how packets are routed from source to destination
- It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.
- Routers are the layer 3 devices, they are specified in this layer and used to provide the routing services within an internetwork.
- The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6

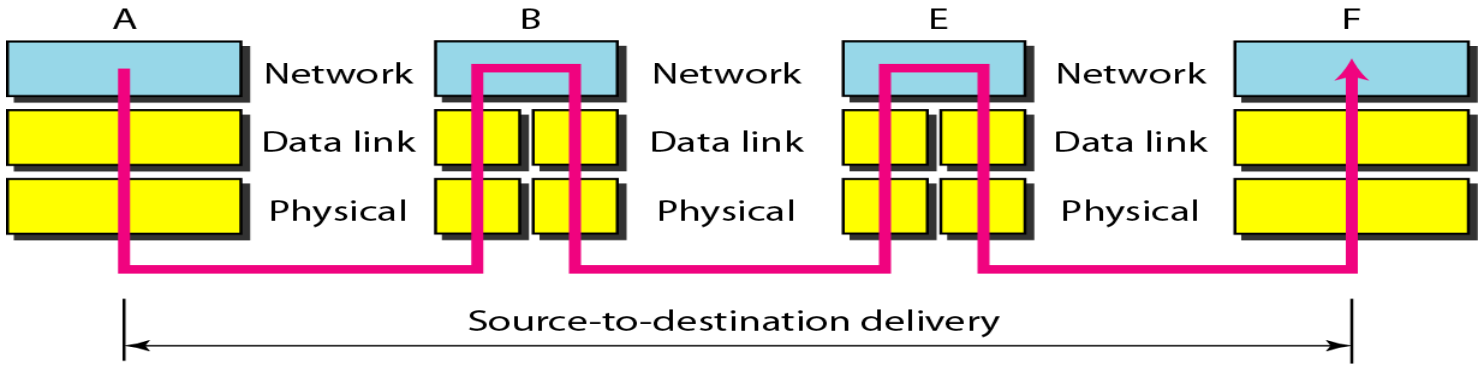
Network Layer

- If too many packets are present in the subnet at the same time, they will get in one another's way, forming bottlenecks.
- Handling congestion is also a responsibility of the network layer
- More generally, the quality of service provided (delay, transit time, jitter, etc.) is also a network layer issue
- When a packet has to travel from one network to another to get to its destination, many problems can arise. The addressing used by the second network may be different from that used by the first one.
- The second one may not accept the packet at all because it is too large. The protocols may differ, and so on
- It is up to the network layer to overcome all these problems to allow heterogeneous networks to be interconnected.
- In broadcast networks, the routing problem is simple, so the network layer is often thin or even nonexistent.

Network Layer

- Source - to – Destination Delivery





Transport Layer

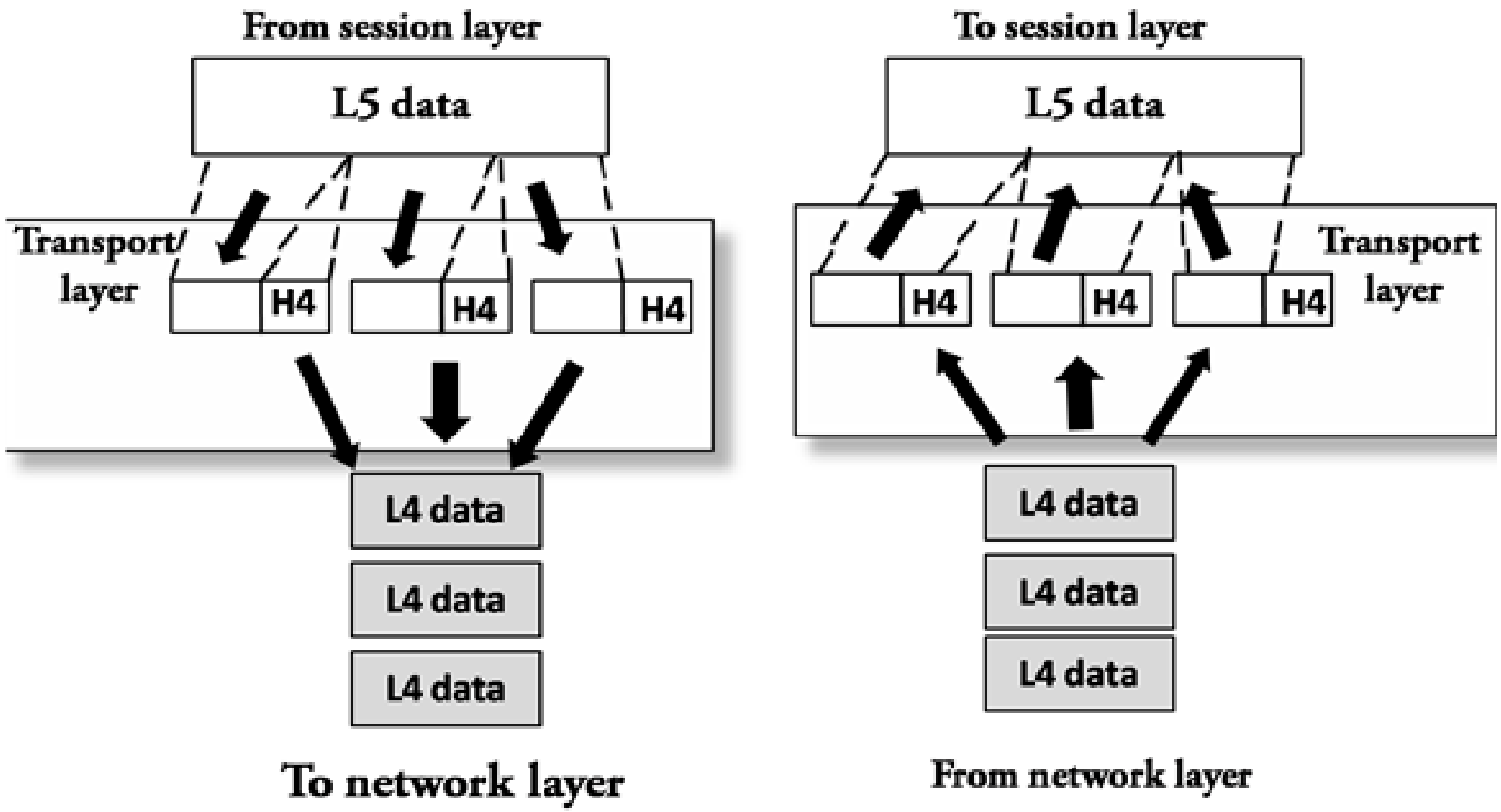
Segmentation

Flow Control

Error Control

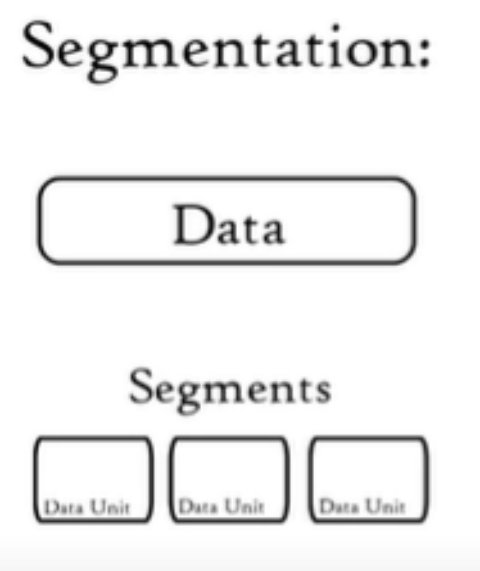
Connection and Connectionless Tx

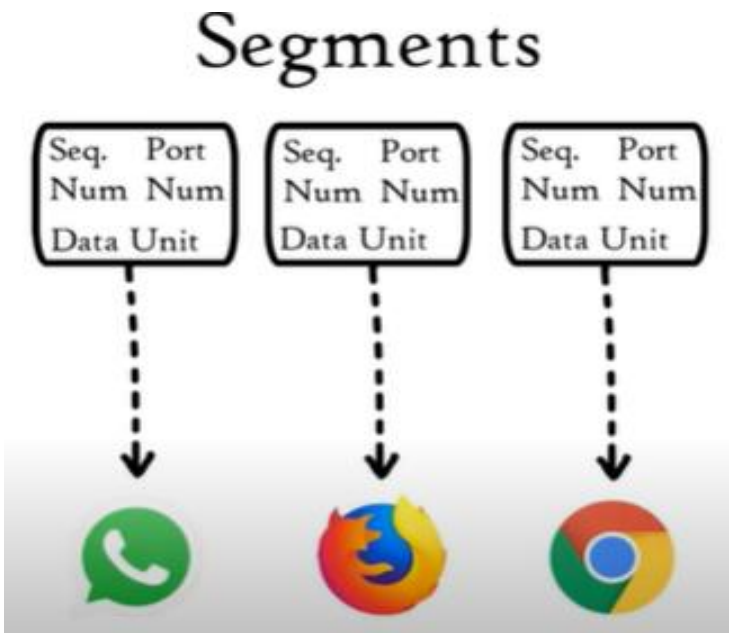
Transport Layer



Transport Layer

- Each segment contain source and destination port number and sequence number
- Port number helps to direct each segment to correct application and sequence number helps to reassemble segments





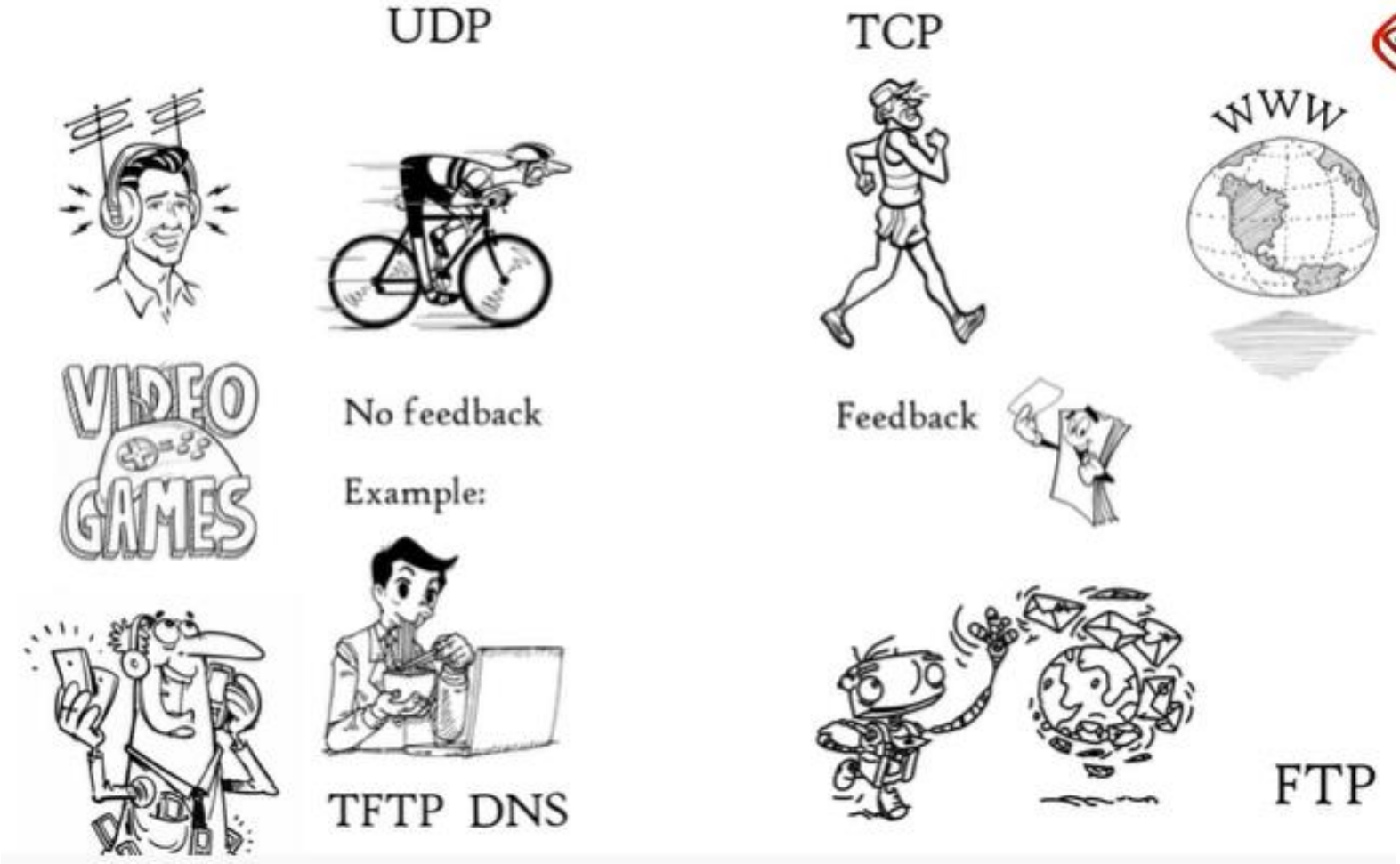
Transport Layer

- The basic function of the **transport layer** is to accept data from above it, split it up into smaller units if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end
- It receives the data from the upper layer and converts them into smaller units known as segments
- The transport layer is a true end-to-end layer; it carries data all the way from the source to the destination.
- A program on the source machine carries on a conversation with a similar program on the destination machine, using the message headers and control messages
- In the lower layers, each protocols is between a machine and its immediate neighbors, and not between the ultimate source and destination machines, which may be separated by many routers
- layers 1 through 3, which are chained, and layers 4 through 7, which are end-to-end,

Transport Layer

- **The two protocols used in this layer are:**
- **Transmission Control Protocol**
  - It is a standard protocol that allows the systems to communicate over the internet.
  - It establishes and maintains a connection between hosts.
  - When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination. The transmission control protocol reorders the packets in the correct order at the receiving end.
- **User Datagram Protocol**
  - User Datagram Protocol is a transport layer protocol.
  - It is an unreliable transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.

Transport Layer



Transport layer

- The transport layer is responsible for the delivery of a message from one process to another
  - Service-point addressing
  - Segmentation and reassembly
  - Connection control
  - Flow control
  - Error control

Transport Layer

- Functions of Transport Layer:
- **Service-point addressing:** Computers run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or port address. The responsibility of the network layer is to transmit the data from one computer to another computer and the responsibility of the transport layer is to transmit the message to the correct process.

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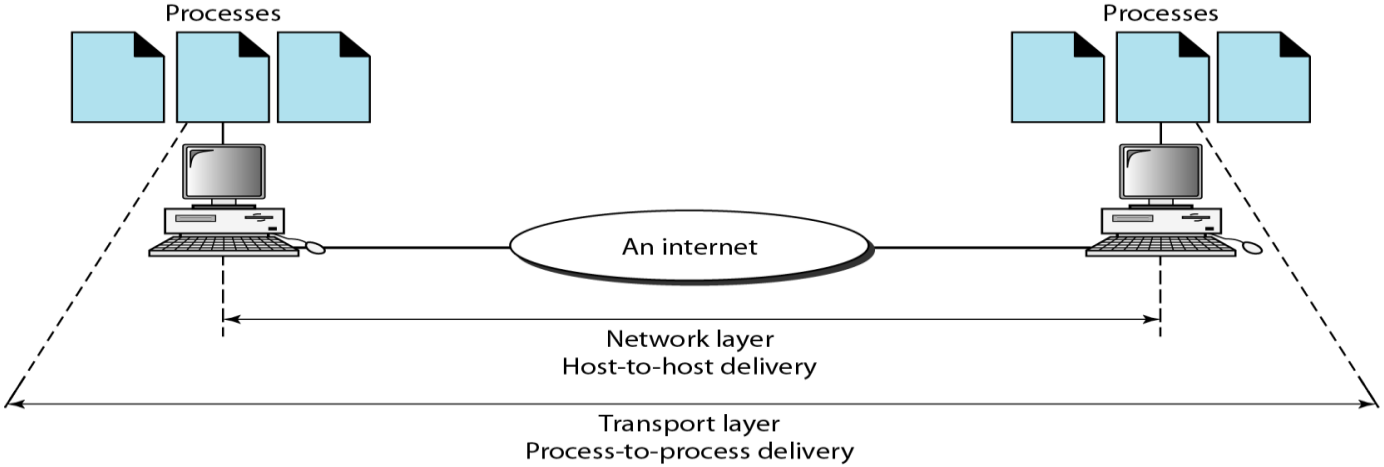
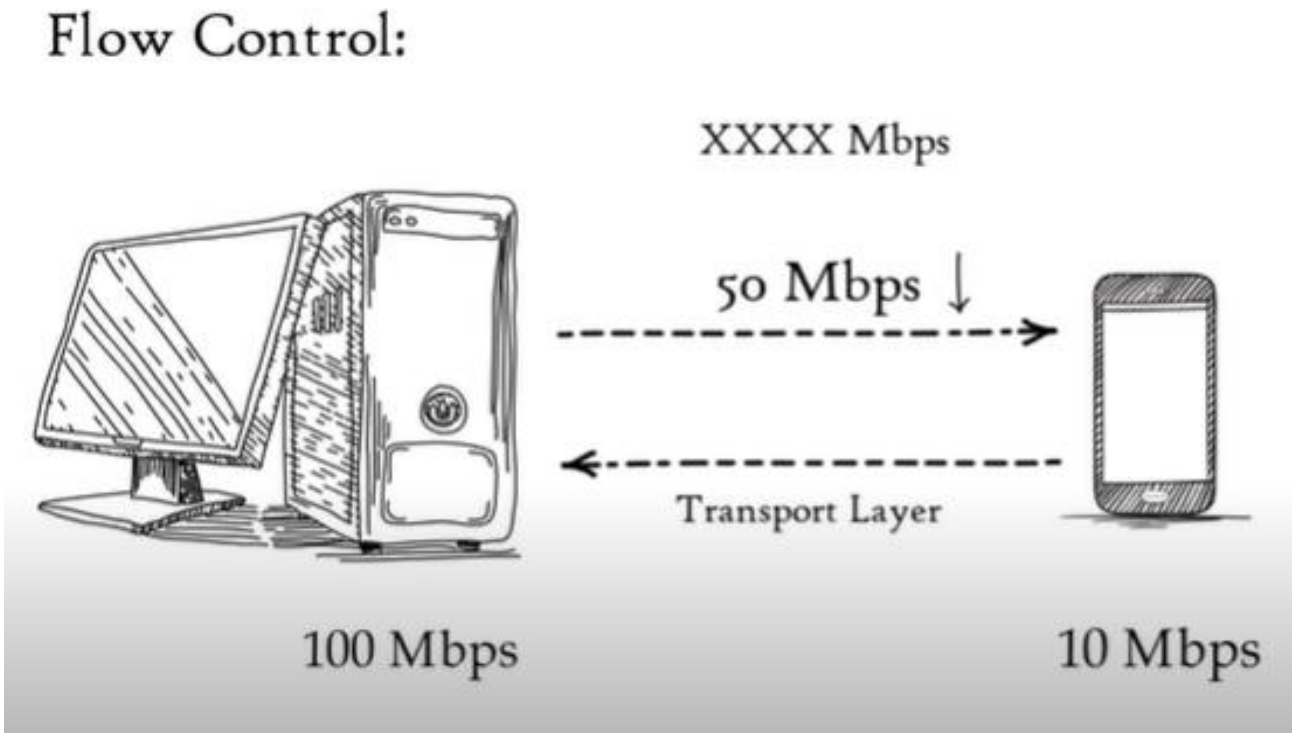


Transport Layer

- **Segmentation and reassembly:** When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.
- **Connection control:** Transport layer provides two services Connection-oriented service and connectionless service. A connectionless service treats each segment as an individual packet, and they all travel in different routes to reach the destination. A connection-oriented service makes a connection with the transport layer at the destination machine before delivering the packets. In connection-oriented service, all the packets travel in the single route.

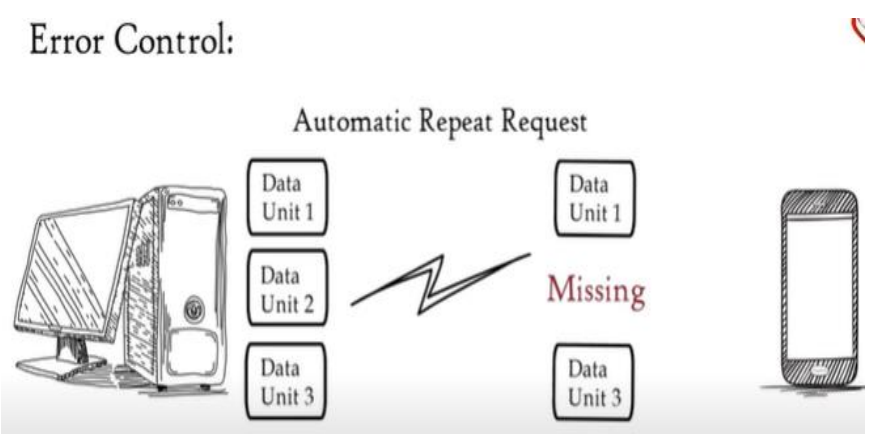
Transport Layer

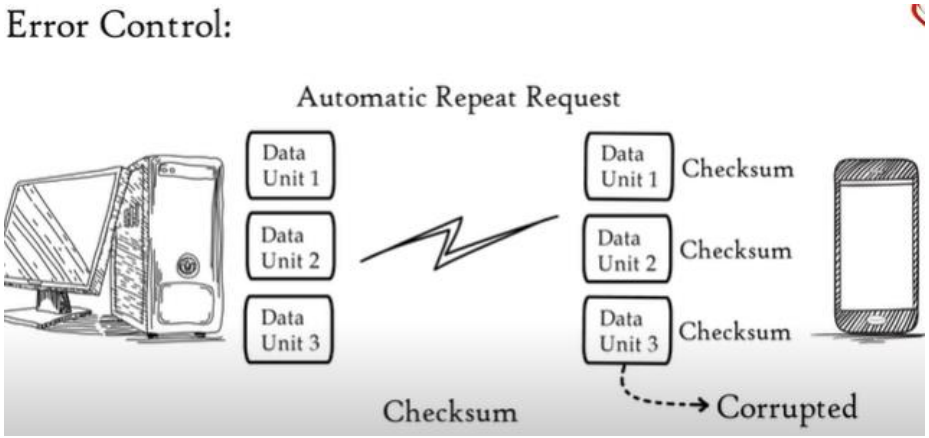
- **Flow control:** The transport layer also responsible for flow control but it is performed end-to-end rather than across a single link.



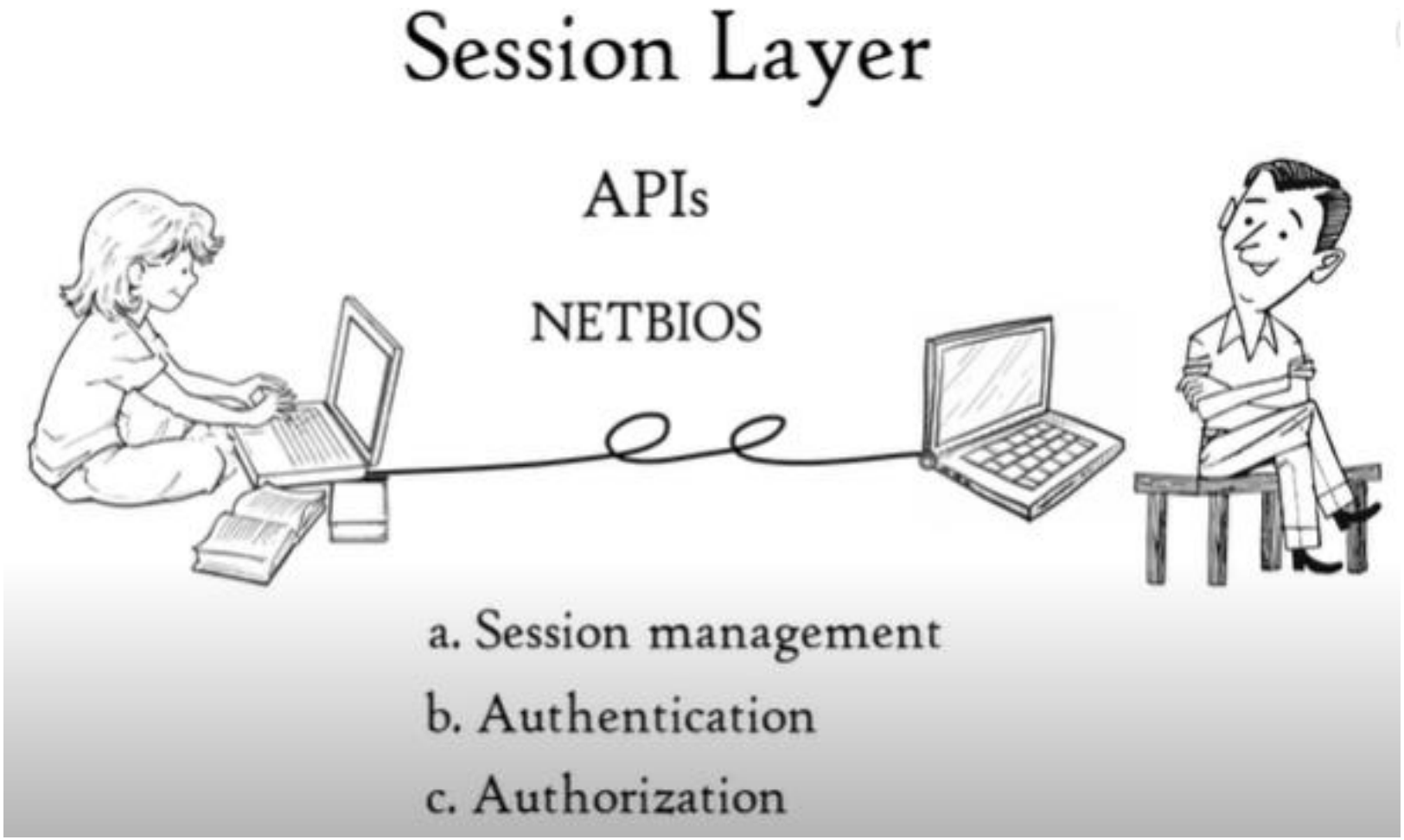
Transport Layer

- **Error control:** The transport layer is also responsible for Error control. Error control is performed end-to-end rather than across the single link. The sender transport layer ensures that message reach at the destination without any error.



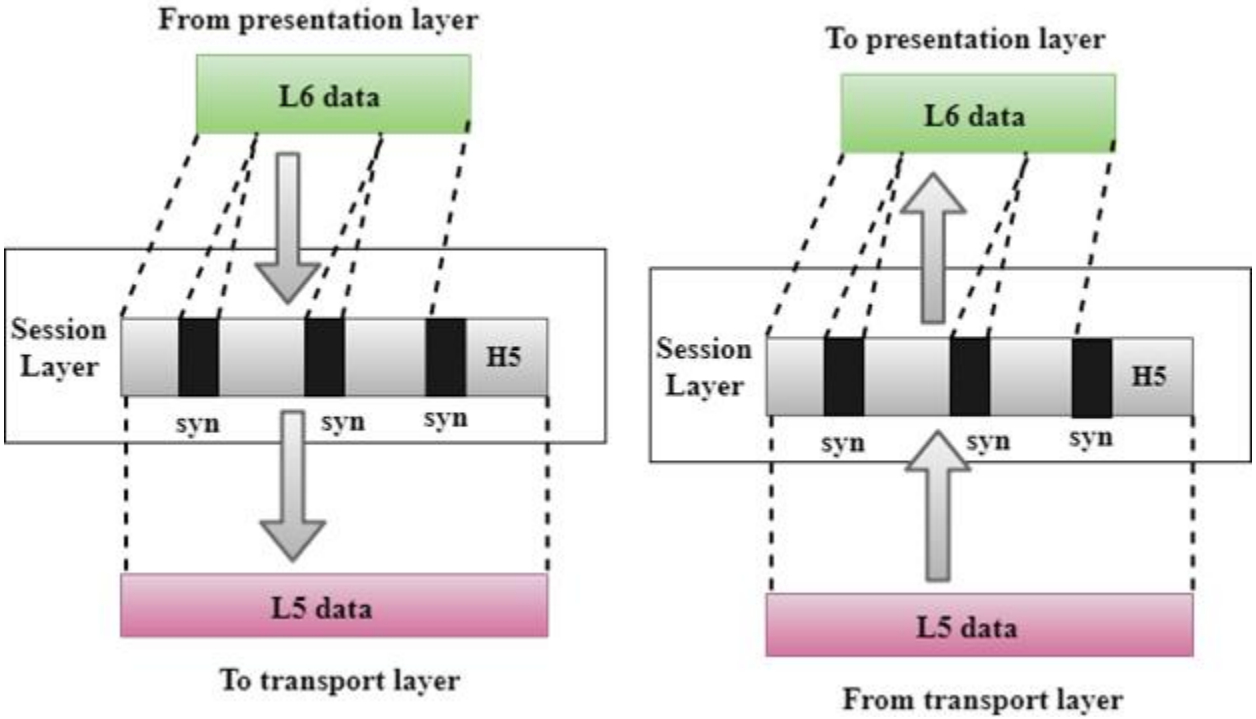


Session Layer



API- Application Programming Interface NETBIOS – Network Basic Input / Output System

Session Layer

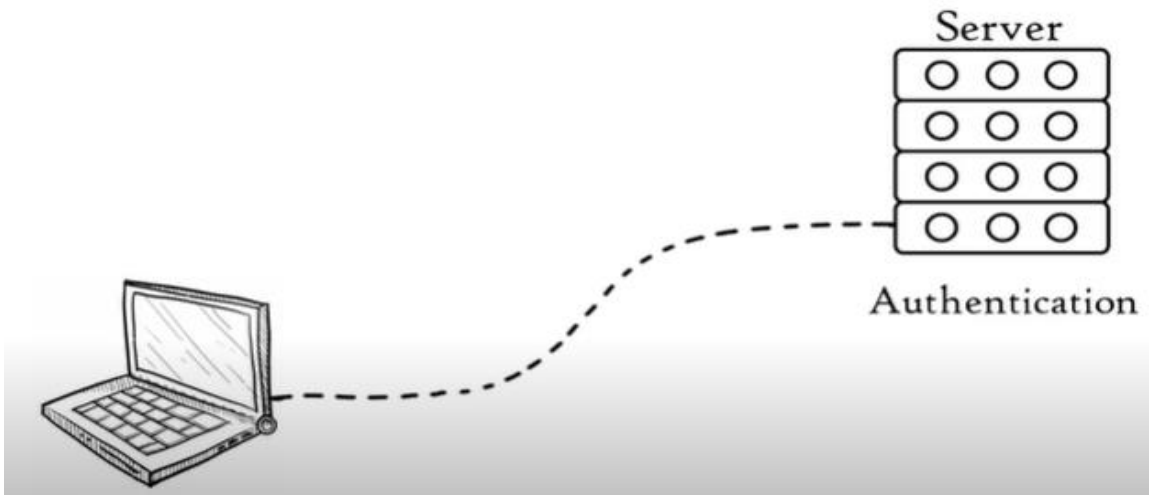


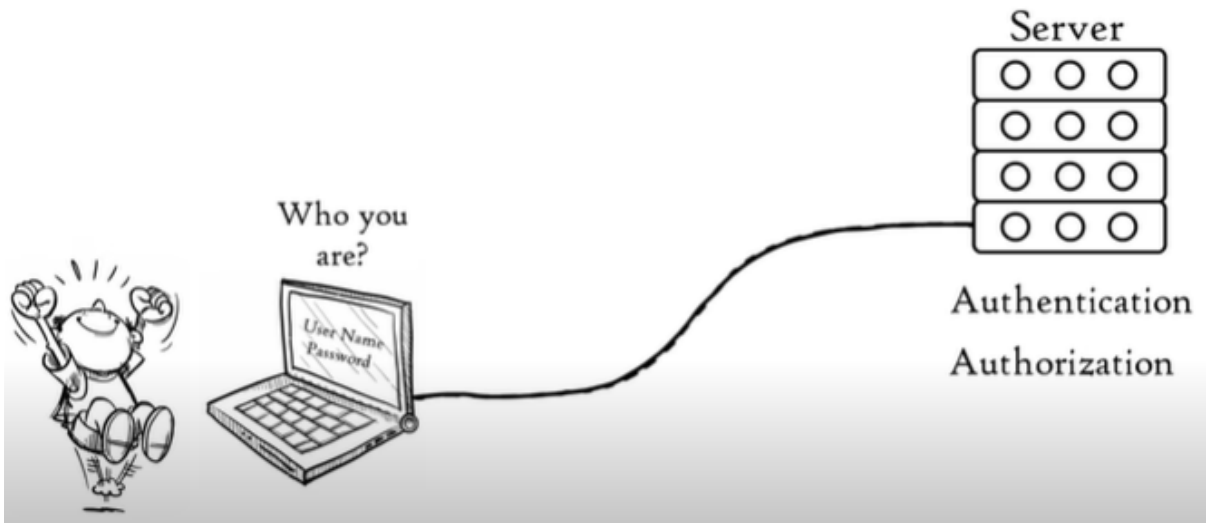
Session Layer

- The session layer allows users on different machines to establish **sessions** between them.
- Sessions offer various services, including
- **Dialog control -** keeping track of whose turn it is to transmit
- **Token management -** preventing two parties from attempting the same critical operation simultaneously
- **Synchronization -** check-pointing long transmissions to allow them to pick up from where they left off in the event of a crash and subsequent recovery

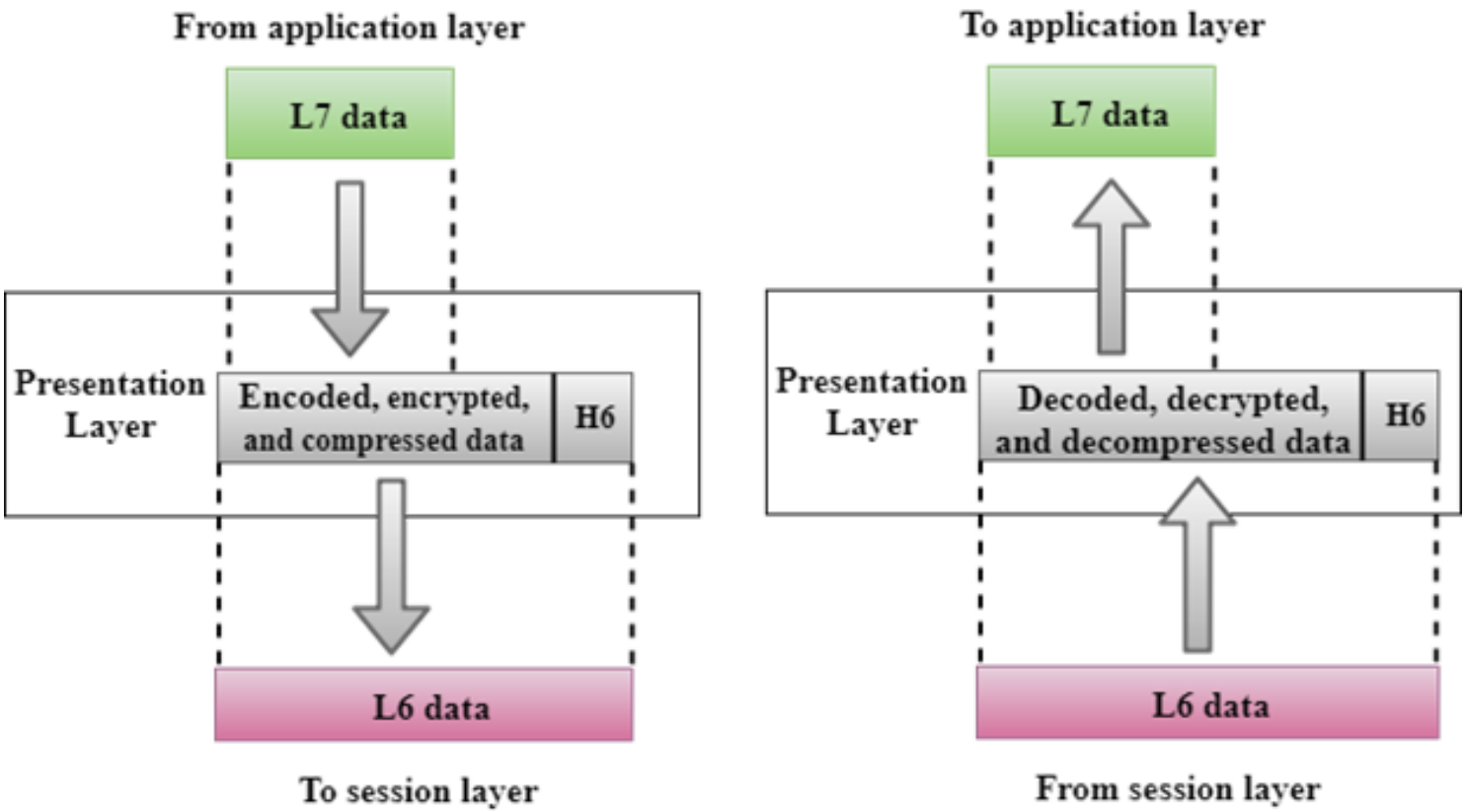
  Example: If a system is sending a file of 800 pages, adding checkpoints after every 50 pages is recommended. This ensures that 50 page unit is successfully received and acknowledged. This is beneficial at the time of crash as if a crash happens at page number 110; there is no need to retransmit 1 to100 pages

Session Layer

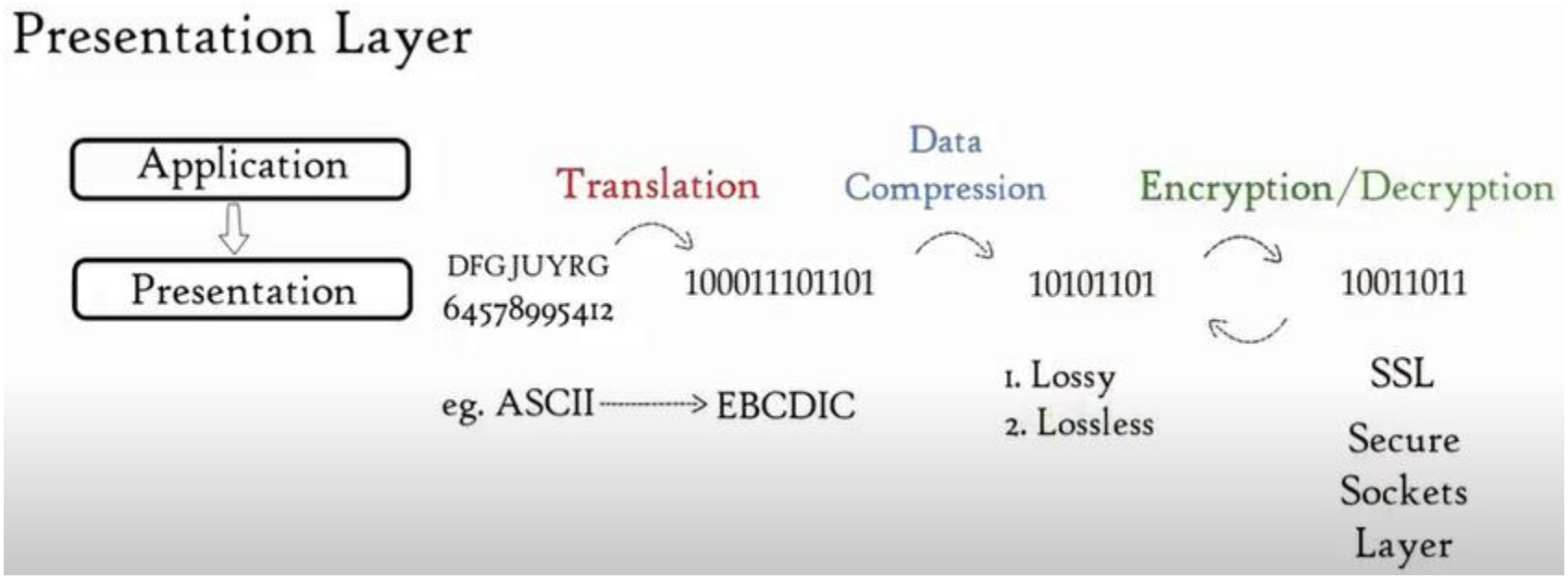




Presentation Layer



OSI - Presentation layer

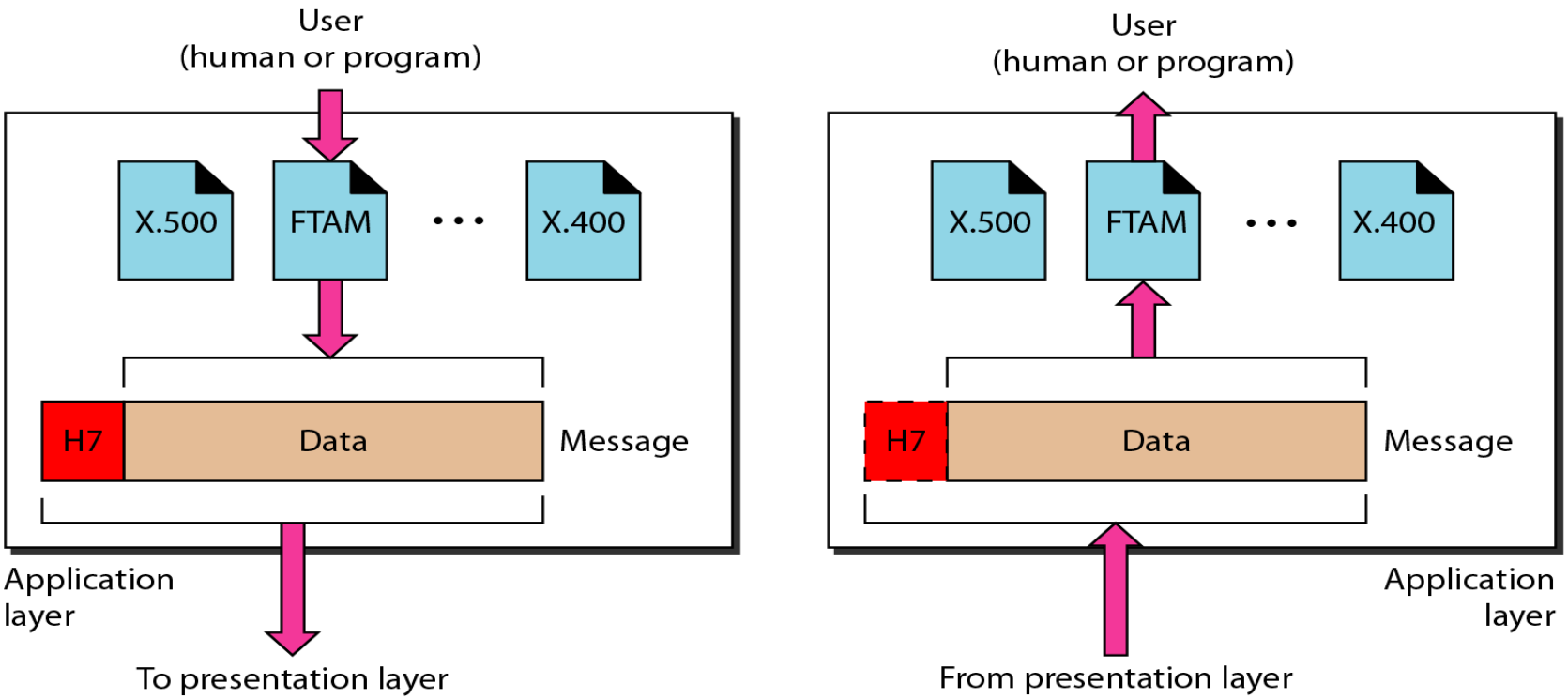


Presentation Layer

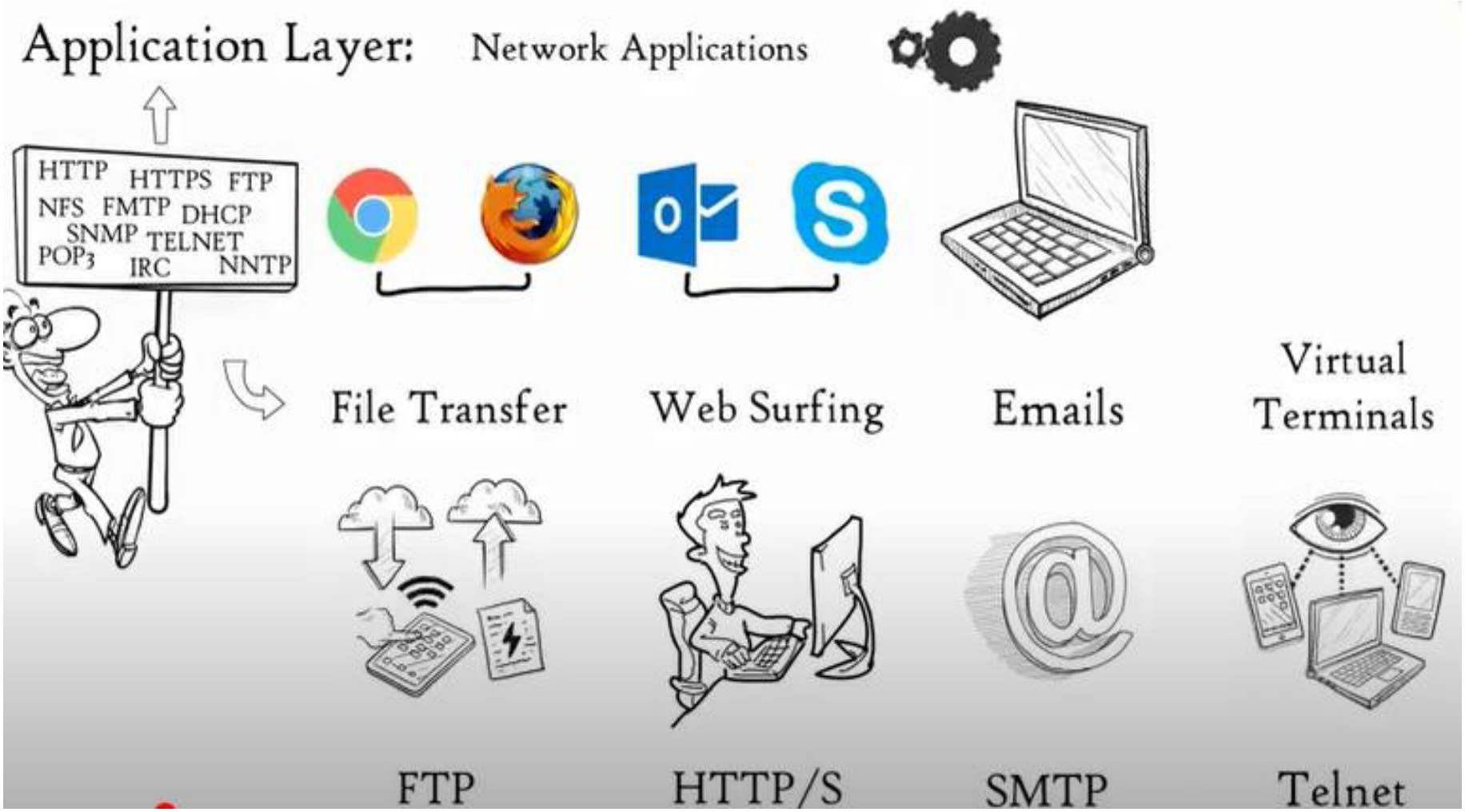
- Functions of Presentation layer:
- **Translation:** The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.
- **Encryption:** Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.
- **Compression:** Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video

OSI - Application layer

- The application layer is responsible for providing services to the user



OSI - Application layer



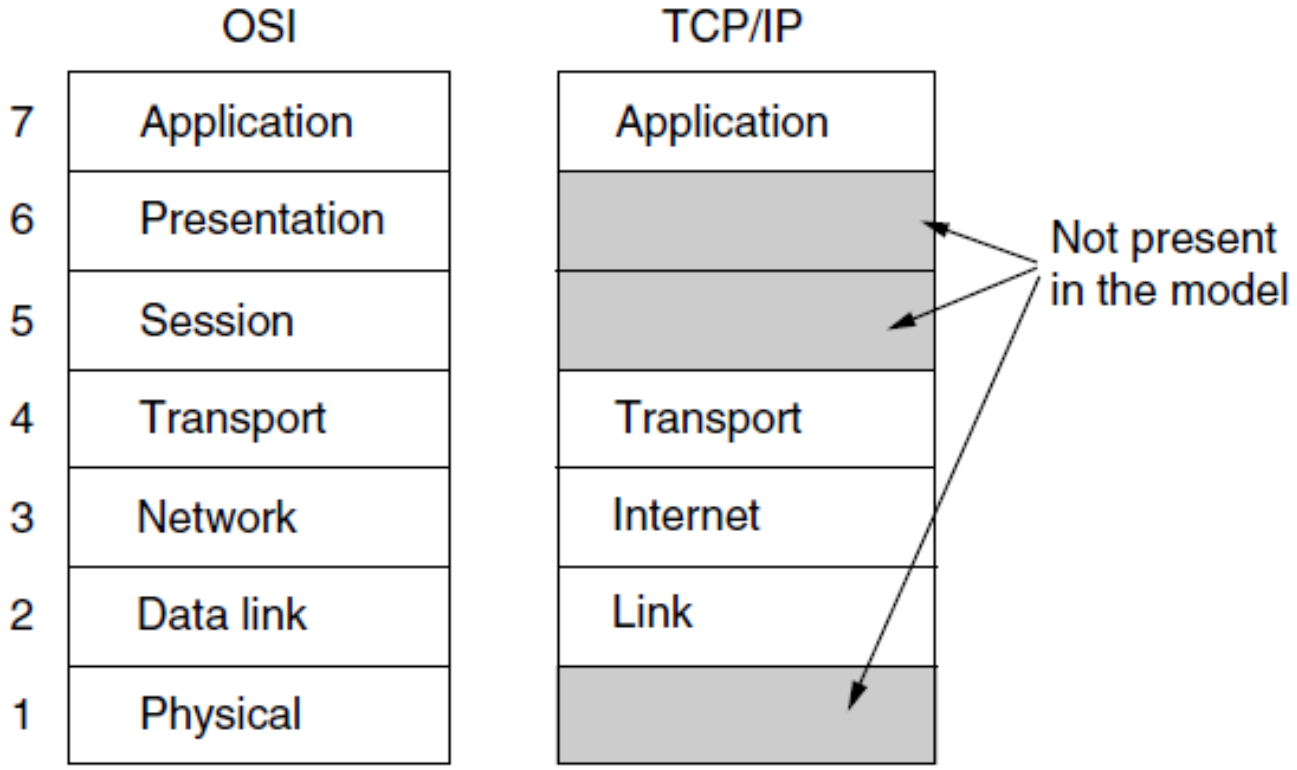
OSI - Application layer

- It is the top most layer of OSI Model
- Enables user or software to get access to the network
- Some services provided by this layer includes: E-Mail, transferring files, distributing the results to user, directory services, network resources, etc.
- The Application Layer contains a variety of protocols that are commonly needed by users
- One widely-used application protocol is **HTTP (Hyper Text Transfer Protocol)**, which is the basis for the World Wide Web. When a browser wants a web page, it sends the name of the page it wants to the server using HTTP. The server then sends the page back.
- Other Application protocols that are used are: File Transfer Protocol**(FTP)**, Trivial File Transfer Protocol**(TFTP)**, Simple Mail Transfer Protocol**(SMTP)**, **TELNET**, Domain Name System**(DNS)** File Transfer Access and Management (**FTAM**) etc.

TCP/ IP Reference Model

- TCP/IP means Transmission Control Protocol and Internet Protocol
- The ARPANET (Advanced Research Project Agency Network) was a research network sponsored by U.S. Department of Defense
- It eventually connected hundreds of universities and government installations, using leased telephone lines.
- When satellite and radio networks were added later, the existing protocols had trouble interworking with them, so a new reference architecture was needed.
- Ability to connect multiple networks in a seamless way was one of the major design goals.
- This architecture later became known as the **TCP/IP Reference Model**

TCP/ IP Reference Model



The TCP/IP reference model.

TCP/ IP

- **Protocols** are set of rules which govern every possible communication over a network
- These protocols describe the movement of data between the source and destination or the internet
- They also offer simple naming and addressing schemes

The Link Layer

- Packet-switching network based on a connectionless layer that runs across different networks
- Lowest layer in the model, the **link layer** describes what links such as serial lines and classic Ethernet must do to meet the needs of this connectionless internet layer.
- It is not really a layer, but rather an interface between hosts and transmission links.

TCP/ IP

The Internet Layer

- The **internet layer** is the linchpin that holds the whole architecture together
- It is corresponding roughly to the OSI network layer
- Its job is to permit hosts to inject packets into any network and have them travel independently to the destination
- They may even arrive in a completely different order than they were sent, in which case it is the job of higher layers to rearrange them, if in-order delivery is desired
- Note that ‘‘internet’’ is used here in a generic sense, even though this layer is present in the Internet.

TCP/ IP

- The internet layer defines an official packet format and protocol called **IP** (Internet Protocol), plus a companion protocol called **ICMP** (Internet Control Message Protocol) that helps it function
- The job of the internet layer is to deliver IP packets where they are supposed to go
- Packet routing is clearly a major issue here, as is congestion

TCP/ IP

The Transport Layer

- Two end-to-end transport protocols have been defined here
- The first one, **TCP** (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet.
- It segments the incoming byte stream into discrete messages and passes each one on to the internet layer
- At the destination, the receiving TCP process reassembles the received messages into the output stream
- TCP also handles flow control to make sure a fast sender cannot swamp a slow receiver with more messages than it can handle

TCP/ IP

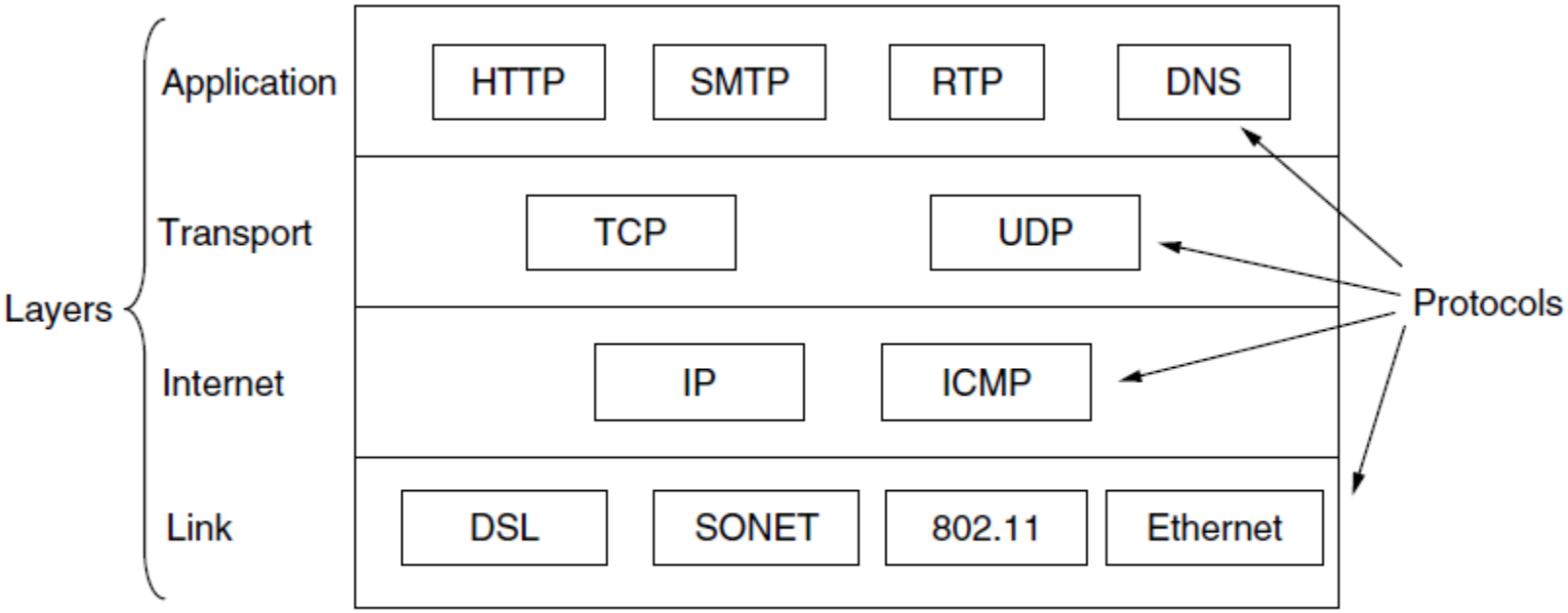
- The second protocol in this layer, **UDP (User Datagram Protocol),** is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own
- It is also widely used for one-shot, client-server-type request-reply queries and applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video

TCP/ IP

The Application Layer

- The TCP/IP model does not have session or presentation layers.
- Instead, applications simply include any session and presentation functions that they require
- It contains all the higher- level protocols
- virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP) , Domain Name System (DNS), for mapping host names onto their network addresses, HTTP, the protocol for fetching pages on the World Wide Web, and RTP, the protocol for delivering real-time media such as voice or movies

TCP/ IP



The TCP/IP model with some protocols

TCP/ IP

- Hybrid Model

5	Application
4	Transport
3	Network
2	Link
1	Physical